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Effects of droughts and warm months on phosphorus and nitrogen concentrations in surface waters globally

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Effects of droughts and warm months on phosphorus and nitrogen concentrations in surface waters globally

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Abstract:

Phosphorus (P) and nitrogen (N) concentrations in surface waters are substantially affected by droughts and warm months. However, previous studies of phosphorus and nitrogen concentrations under these hydro-climatic extremes are limited to a small number of regions, mainly in parts of North America, Europe and Australia. Here we perform the first global-scale analysis of phosphorus and nitrogen concentrations in surface waters under droughts, warm months and compound drought-warm months for the period 1980-2010. To achieve this, we developed an approach to downscale existing simulation data of total phosphorus (TP) and total nitrogen (TN) concentrations from annual to monthly level with machine learning at 0.5 x 0.5 degree resolution. The results indicate generally mixed responses in phosphorus and nitrogen concentrations under drought, with some regions showing increases in the concentrations of phosphorus and nitrogen due to drought and others showing decreases. Additionally, the results show more consistent increases in phosphorus (P) concentrations during warm months (average change: +21%) and compound drought-warm months (average change: +10%), which was mainly attributed to less dilution during periods of low flow and high temperature. Furthermore, there is a general decrease in the N:P ratio observed globally during warm months (average change: −11%). The results from this study provide a broader understanding of the effects of droughts and warm months on phosphorus and nitrogen in surface waters compared to previous local-scale studies. Furthermore, the results highlight the significant changes in phosphorus and nitrogen ratios during droughts and warm months, which may significantly influence freshwater ecosystem health worldwide.

1. Introduction

Phosphorus (P) and nitrogen (N) are key water quality constituents that strongly influence freshwater ecosystem health (Zhou et al., 2024; Kanter and Brownlie, 2019; Azevedo et al., 2015). P and N play an important role in the biogeochemistry of rivers and are crucial to support aquatic life in surface waters globally (Dodds and Smith, 2016). However, when the concentrations of phosphorus and nitrogen become elevated in freshwater bodies, this can lead to eutrophication with associated excessive algal growth, depletion of dissolved oxygen and harmful effects for aquatic species (Li et al., 2017; Suresh et al., 2023). The issue of eutrophication is widespread in many of the world's surface waters and can persist until the sources of phosphorus and nitrogen are mitigated by, for instance, improving pollution regulation (Álvarez et al., 2017). Additionally, long-term nutrient pollution can cause legacy effects from, for instance, highly polluted groundwater which can be a major source of nitrogen in surface waters (Barcala et al., 2021; Liu et al., 2024).

The concentrations of phosphorus and nitrogen in rivers and lakes are driven by both hydro-climatic factors such as water level, water temperature and precipitation as well as socio-economic factors including land use and wastewater treatment levels and capacity (Li et al., 2024; van Vliet et al., 2023). In addition, the responses of phosphorus and nitrogen in surface waters depend on the types of pollution source, including point sources or diffuse sources (Mosley, 2015; Graham et al., 2024). For instance, in regions with consistent point sources of nutrient pollution, periods of low discharge may lead to increased concentrations of phosphorus or nitrogen due to reduced dilution (Sprague, 2005; Macintosh et al., 2011; Hrdinka et al., 2012). Whereas, in regions with mostly diffuse sources of pollution, periods of drought reduce the mobilization and transport of nutrients from land to surface waters leading to decreased phosphorus and nitrogen loads to surface waters (Oelsner et al., 2007, Ziellnski et al., 2009, Baurès et al., 2013). Nevertheless, for each river or lake system, the concentrations of phosphorus and nitrogen are driven by a complex interplay of nutrient form, biogeochemistry, climate and atmospheric deposition, hydrology, land cover and use, water management, and infrastructure (Li et al., 2024; Payen et al., 2021; Smith et al., 2006). Climate change, which is expected to increase the frequency of droughts and heatwaves in many regions of the world (Jeppesen et al., 2010), may lead to increasing risks for algae blooms, reduced dilution in rivers and lakes and increasing water temperatures, which may increase low oxygen events (Graham et al., 2025; van Vliet et al., 2023). Periods of droughts and high temperatures can also affect the N:P ratio in surface waters, which is an important stoichiometric indicator in freshwater ecosystems (Beusen et al., 2016). Furthermore, socio-economic development such as population growth and land use change can contribute to the increased risks of eutrophication by increasing the concentrations of phosphorus and nitrogen in surface waters (Hilton et al., 2006; Nijboer and Verdonschot, 2004).

Most of the existing studies related to drought impacts on nutrient concentrations focus on local or regional scale (Mosley, 2015). Furthermore, these studies are often located in either North America, Europe or Australia due to more widely available water quality monitoring data in these regions compared to other regions (e.g. Africa and central Asia) (van Vliet et al., 2023). For example, impacts of drought on phosphorus and nitrogen have been studied in Rio Grande (Oelsner et al., 2007), South Platte (Sprague, 2005), Meuse (van Vliet and Zwolsman, 2008) and the Murray-Darling (Mosley et al., 2012) river basins. Furthermore, the impacts of compound drought-warm events on nutrient concentrations have been studied in the Rhine river basin, Europe (Zwolsman and van Bokhoven, 2007). These local-scale studies provide an important basis for understanding the driving mechanisms of nutrient concentrations during periods of drought and high temperature. However, local and regional (river basin) studies do not provide globally applicable understanding and consistent picture of the effects of droughts and compound drought-warm events on phosphorus and nitrogen concentrations.

The closest studies to incorporate a global-scale study of the impacts of droughts on nutrient concentrations are Mosley (2015), van Vliet et al. (2023) and Graham et al. (2024). For instance, Mosley (2015) and van Vliet et al. (2023) reviewed multiple local and regional studies to show consistencies in the responses of nutrients during hydro-climatic extremes across different regions. Furthermore, Graham et al. (2024) performed data analyses to quantify responses of total phosphorus (TP) and total nitrogen (TN) concentrations during droughts, heatwaves and compound events for approximately 50,000 water quality monitoring stations across the world. Nevertheless, there remain significant portions of the world where the response of phosphorus and nitrogen concentrations under these hydro-climatic extremes have not been fully studied or remain poorly understood.

To study the effects of droughts and warm events on nutrient concentrations at the global level, large-scale nutrient models could potentially be used. These consider the effects of different nutrient sources, such as agricultural runoff or human and industrial wastewater effluent, and they then simulate the physical and biogeochemical processes affecting phosphorus and nitrogen, such as advection, dilution and denitrification. Examples of global-scale nutrient models include the Global NEWS model (Mayorga et al., 2010), the MARINA model (Strokal et al., 2016) and IMAGE-GNM (Beusen et al., 2015). However, the temporal resolution of these models is typically too coarse to capture the impacts on hydro-climatic extremes on phosphorus and nitrogen. For instance, most of the existing large-scale nutrient models operate at an annual (yearly) resolution or do not capture dynamic changes, whereas drought events and identification methods are commonly applied at monthly to daily resolution. Additionally, existing global nutrient models are not specifically designed to capture the impacts of interannual extremes, but rather to capture the overall yearly average dynamics of phosphorus and nitrogen. To address this shortcoming, one option is to combine

traditional process-based water quality models with data-driven approaches. For instance, machine learning methods could be applied to both temporally downscale and improve the predictive capabilities of global nutrient models during hydro-climatic extremes (Wang et al., 2020; Zhi et al., 2024; Tang et al., 2026). Such an approach could enable a global-scale study on the effects of droughts on phosphorus and nitrogen concentrations.

In this study, we quantify the impacts of droughts , warm months and compound drought-warm months on total phosphorus (TP) and total nitrogen (TN) concentrations globally over the period 1980-2010. To achieve this, we developed a hybrid model framework combining annual modelled TP and TN concentrations from the Integrated Model to Assess the Global Environment - Global Nutrient Model (IMAGE-GNM) (Beusen et al., 2015) with machine learning techniques (random forest) to downscale yearly average TP and TN simulated concentration to monthly time series with the goal to improve the model performance on a monthly basis and capture impacts during droughts and warm months. Furthermore, we quantified the main drivers of TP and TN concentration changes during periods of drought, warm months and compound events.

2. Methods

2.1 Hybrid nutrient modelling framework

We developed a hybrid process-based and machine learning modelling approach to simulate total phosphorus (TP) and total nitrogen (TN) concentrations at the monthly level in surface waters globally (Extended Data Fig. 1). These monthly simulations were then used to estimate the impacts of drought and warm months with high water temperature on global nutrient concentrations over the 1980-2010 period. The process-based simulations of TP and TN were obtained from existing datasets simulated at the annual level by the IMAGE-GNM global nutrient model (Beusen et al., 2022) on 0.5° x 0.5° spatial resolution for the period of 1980-2010. We used random forest as machine learning technique to downscale the annual level TP and TN concentrations to the monthly level series. A full model description of the IMAGE-GNM nutrient model can be found in Beusen et al. (2015). The precipitation and air temperature inputs were obtained from the ECMWF ERA20C reanalysis (Poli et al., 2016) with monthly observations of the CRU TS 3.2.3, which are aligned with observations over the historic period. Additionally, hydrological variables including river discharge were simulated by the global hydrological model PCR-GLOBWB (van Beek et al., 2011) and water temperatures were simulated at the global level by solving the surface water energy balance (van Beek et al., 2012) on a 0.5° x 0.5° spatial resolution.

We applied random forest (RF) to downscale the annual TP and TN concentration simulations from IMAGE-GNM to the monthly level on a 0.5° x 0.5° spatial resolution. This involves using

four key variables as the predictors (i.e., monthly discharge, monthly water temperature, monthly precipitation and the annual TP or TN concentrations from IMAGE-GNM) (Eqs. 1, 2). Additionally, the target variable was either chosen as the monthly TP concentration or the monthly TN concentration (Eqs. 1, 2).

$$TP_{month,t} = RF\ (TP_{year,t}, Q_{month,t}, Tw_{month,t}, Precip_{month,t}) \quad 1.$$

Where $TP_{month,t}$ is the monthly averaged total phosphorus concentration (mg P l^{-1}) at time t, RF is Random Forest model, $TP_{year,t}$ is the annual total phosphorus concentration (mg P l^{-1}) at time t from IMAGE-GNM, $Q_{month,t}$ is the monthly discharge (km^3 $year^{-1}$) at time t, $Tw_{month,t}$ is the monthly water temperature (^oC) at time t, $Precip_{month,t}$ is the monthly precipitation (m) at time t.

$$TN_{month,t} = RF\ (TN_{year,t}, Q_{month,t}, Tw_{month,t}, Precip_{month,t}) \quad 2.$$

Where $TN_{month,t}$ is the monthly averaged total nitrogen concentration (mg N l^{-1}) at time t, RF is Random Forest model, $TN_{year,t}$ is the annual total nitrogen concentration (mg N l^{-1}) at time t from IMAGE-GNM, $Q_{month,t}$ is the monthly discharge (km^3 $year^{-1}$) at time t, $Tw_{month,t}$ is the monthly water temperature (^oC) at time t, $Precip_{month,t}$ is the monthly precipitation (m) at time t.

Our choice of predictors is driven by two main factors. Firstly, we aim to include the key variables and mechanisms which affect TP and TN concentrations at both the annual and monthly levels, and secondly we aim to avoid overfitting the model which may result if we included too many predictors. The relative amounts of diffuse and point source pollution are fully represented in the process-based (annual) simulations of TP and TN from the IMAGE-GNM model (Beusen et al., 2015; Beusen et al., 2022), which are provided to the random forest component. In addition, the influence of groundwater and land use factors (e.g. agricultural drainage) are also incorporated in the process-based (annual) simulations of TP and TN from IMAGE (see Beusen et al., 2015; Beusen et al., 2022). If we were to include additional variables in the random forest related to, for instance, land use, this would effectively be incorporating them twice in the model and potentially lead to overfitting. Therefore, by using the predictors in the current model, this keeps the approach more parsimonious when applying at the global level. The machine learning model is trained to capture concentration-discharge (C-Q) relationships across a wide range of regions globally, in addition to C-Tw and C-Precip relationships. Furthermore, to reduce the effects of spatial and temporal biases in the available monitoring data for training and testing, we used stratified sampling. This ensures that, proportionally speaking, the same amount of TP or TN data is used across different regions of the world and for each decade when training and testing the model. For unmonitored locations, the range of values of the predictors (i.e.

discharge, water temperature, precipitation and annual TP or TN) is assumed to remain within the range observed during the training procedure, which is a key assumption made when applying (hybrid) process-based and machine learning-based methods at the global scale (Zhi et al., 2024; Yan et al., 2025; Graham et al., 2025).'

We then applied a physical constraint to the model such that the amount of mass (or load) of phosphorus and nitrogen was conserved when downscaling from annual to monthly level. To achieve this, we firstly obtained estimates of the monthly TP and TN loads by multiplying the monthly concentrations obtained with machine learning (i.e. RF) by the monthly discharge simulated by PCR-GLOBWB. We then applied a correction factor on monthly time step to ensure that the monthly summed up loads were consistent with the loads from the annual timestep simulations of TP and TN (Eqs. 3, 4). The correction factor ($C_{year,t}$) is calculated as the yearly TP and TN loads of IMAGE-GNM divided by the sum of the monthly loads of TP and TN of the RF method for each grid cell. We then divided the corrected loads by the discharge again from PCR-GLOBWB to reobtain corrected monthly estimates of TP and TN concentrations.

$$TP'_{month,t} = C_{year,t}\, TP_{month,t} \quad 3.$$

Where $TP'_{month,t}$ is the mass balance-corrected monthly total phosphorus load (mg P) at time t, $C_{year,t}$ is a correction factor to ensure the total phosphorus loads at the monthly level add up to the same loads at annual level based on IMAGE-GNM for each grid cell. The correction factor for TP is the ratio of the annual load simulated by IMAGE-GNM and the estimated annual load based on the simulations of TP from Eq. 1. $TP_{month,t}$ is the uncorrected monthly total phosphorus load (mg P) at time t.

$$TN'_{month,t} = C_{year,t}\, TN_{month,t} \quad 4.$$

Where $TN'_{month,t}$ is the mass balance-corrected monthly total nitrogen load (mg N) at time t, $C_{year,t}$ is a correction factor to ensure the total nitrogen loads at the monthly level add up to the same loads at annual level based on IMAGE-GNM for each grid cell. The correction factor for TN is the ratio of the annual load simulated by IMAGE-GNM and the estimated annual load based on the simulations of TN from Eq. 2. $TN_{month,t}$ is the uncorrected monthly total nitrogen load (mg N) at time t.

The model training and testing was carried out on 1,091,746 and 272,941 monthly observations for TP concentration (an 80/20% split), respectively (Fig. 1a) (Graham et al., 2024). Additionally, the model training and testing used 184,748 and 46,187 monthly observations for TN concentration (80/20% split), respectively (Fig. 1b) (Graham et al., 2024). These monthly data were obtained firstly by averaging the available number of TP or TN

measurements for each month at each location. The total number of TP and TN observations varies per station, with some locations providing only several measurements while others provide more continuous time-series of TP and TN measurements over the training period (1980-2010) (see Extended Data Fig. 2,3). We used stratified random sampling across 356 river basins worldwide (based on HydroBASINS level 4, shown in Extended Data Fig. 3a-b) and for each decade between 1980-2010 when training and testing the Random Forest model. This ensures that, proportionally speaking, the same amount of data is used for training and testing across different regions of the world and over time, which reduces the effects of potential spatial and temporal biases in the training data. Additionally, we used 100 trees when downscaling the TP and TN concentrations from annual to monthly level.

2.2 Identification of droughts, warm months and compound events

To identify the periods of drought, we used the variable threshold level method (Sarailidis et al., 2019; van Huijgevoort et al., 2012; Sung and Chung, 2014; Wanders et al., 2015). We selected the lower 10th percentile of discharge for each month of the year at the grid cell level during 1980-2010. Then, drought periods were identified as months below these lower thresholds of discharge. We chose to use the variable threshold level method as opposed to alternative drought identification methods based on precipitation such as the Standardised Precipitation Index (SPI), because discharge has a more direct effect on TP and TN concentrations whereas the influence of precipitation is more indirect. For instance, discharge effects the dilution capacity and heat capacity and thus temperature of surface waters, as well as the mobilisation of surface water pollutants, including phosphorus and nitrogen. We further identified warm months based on when the average water temperature is above the upper 90th percentile (per grid cell). Additionally, we identified compound events as months experiencing hydrological drought based on the variable threshold level method (10th percentile discharge) and when the water temperatures were above the 90th percentile at the grid cell level. By using these approaches, this means that droughts and warm months will occur 10% of the time for each grid cell over the period 1980-2010, which is consistent with previous studies which have used similar threshold values (Mukama et al., 2025; Simeone et al., 2024; Heudorfer and Stahl, 2016).

2.3 Model evaluation

We validated the performance of our hybrid nutrient modelling framework both with respect to the performance of IMAGE-GNM for TP and TN simulations and in terms their ability to capture the monthly variations for both non-extreme and extreme conditions (droughts, warm months, compound events). The performances were assessed using the normalised Root-Mean-Squared-Error (nRMSE), which is the main performance metric used during the development of the process-based GNM model (Beusen et al., 2015). The nRMSE,

normalised by the observed mean, is commonly used for validation of global water quality models as it enables a consistent comparison across regions with different mean pollution levels of as well as different levels of variability (Graham et al., 2025; Jones et al., 2023). Additionally, we validated the downscaled simulations of TP and TN at the monthly level using the nRMSE. Values of nRMSE closer to 0 represent a greater performance in terms of the difference between observed and simulated TP and TN concentrations (e.g. Jones et al., 2023; Beusen et al., 2015). The nRMSE values can also be converted into the associated RMSE values (in mg l^{-1}) by multiplying the nRMSE values by the observed mean concentrations of TP or TN for each station. The performance was further assessed during periods of drought, warm months and compound events compared to non-extreme events for 1980-2010 (see also Wanders et al., 2015 and Chen et al., 2026 for previous studies using PCR-GLOBWB and DynWat for the identification of drought and water temperature extremes). Time-series of monthly averaged TP and TN concentrations simulated by the downscaling approach were also directly compared to observational data for selected river basins and regions. These locations were chosen based on the availability of continuous (monthly) data over the study period (1980-2010), as well as to capture the performance of the TP and TN simulations across multiple different regions of the world.

2.4 Analysis of P and N responses under droughts, warm months and compound drought-warm months

We analysed the global patterns and trends in TP and TN concentration during 1980-2010 using the downscaled (monthly) dataset, and we quantified the average change in these nutrient concentrations during periods of drought, warm months, compound events. For instance, we quantified the mean TP and TN concentrations globally for each month of the year between the periods 1980-2010, and we quantified the mean trends in TP and TN as well. These trends are useful to quantify as they can show how well the TP and TN trends are preserved using the downscaled dataset compared to previous studies performed at the annual level (e.g. Beusen et al., 2015; Beusen et al., 2022). Furthermore, to quantify the average change in TP and TN concentrations during periods of drought and warm months, we firstly estimated the mean TP and TN concentrations during non-extreme periods, as well as during extremes. Then, we estimated the percentage change in TP and TN concentrations based on the difference between non-extreme and extreme conditions. This procedure was also performed for the N:P ratios during the period 1980-2010. To further explore the potential drivers of phosphorus and nitrogen during periods of drought and high temperature, we analysed monthly time series of nutrient concentrations throughout 1980-2010 on a global level and for six regions (North America, Europe, Asia, South America, Africa and Australia). Further, we determined the statistical significance of these time series of TP and TN per region over the period 1980-2010 using the Mann-Kendall trend test. Additionally, we quantified the contribution of each predictor in the RF component of the

hybrid framework to assess the influence of each hydrological variable (i.e. river discharge, water temperature, and precipitation) on TP and TN concentrations during droughts and warm months. This was performed using both feature importance analysis and assessing the partial dependence of each predictor on the simulated (monthly) concentrations of total phosphorus and total nitrogen.

3. Results

The monthly simulations of total phosphorus (TP) and total nitrogen (TN) concentrations exhibit an overall acceptable level of performance in terms of Root Mean Squared Error, normalised by the mean (nRMSE), for both average (non-extreme) and extreme hydro-climatic conditions (i.e droughts and warm months) (Fig. 2-4, Extended Data Fig. 4). For instance, the median nRMSE values of the monthly TP and TN concentrations are equivalent to 1.0 and 0.9, respectively (Fig. 2c-d). This level of performance is comparative to the annual simulations obtained from IMAGE-GNM (Fig. 2a-b, Extended Data Fig. 5), and a key reason for this similarity in performance is likely due to the mass balance constraint incorporated in the hybrid process-based and random forest modelling approach. Additionally, the performance of the monthly simulations of TP and TN concentrations varies regionally with median nRMSEs for the monthly TP concentrations of 1.1 (North America), 0.9 (South America), 0.9 (Europe), 1.0 (Africa), 1.1 (Asia) and 1.1 (Australia). Additionally, the median nRMSE for the monthly TN concentrations are 0.9 (North America), 1.3 (South America), 0.5 (Europe), 0.9 (Africa), 0.6 (Asia) and 1.0 (Australia). These performance metrics show a slightly better performance in predicting TN compared to TP. Furthermore, validation of the hybrid approach shows an overall acceptable performance during periods of drought and warm months (i.e. high water temperature) (Fig. 2,3). For instance, the median nRMSE value associated with TP concentrations during periods of drought and high temperature are 1.0 and, 0.9, respectively. In addition, the median nRMSEs for TN during drought and warm months are both 0.9.

The hybrid nutrient modelling framework was applied at the global level over the period 1980-2010, and the resulting patterns and trends in monthly time series of TP and TN concentrations were subsequently analysed. Based on this downscaled dataset, the average monthly TP and TN concentrations during 1980-2010 are generally highest in central Europe, parts of Asia and North America, and the strongest trends in TP and TN concentrations tend to occur in central Europe and Southeast Asia (Fig. 5). For example, the mean monthly TP concentrations are found to be 0.52 mg l^{-1} (Europe), 0.21 mg l^{-1} (Asia) and 0.28 mg l^{-1} (North America) over the period 1980-2010 based on the downscaled dataset (Fig. 5a). Furthermore, the mean monthly TN concentrations in these regions are 3.7 mg l^{-1} (Europe), 1.6 mg l^{-1} (Asia) and 1.8 mg l^{-1} (North America) (Fig. 5c) using the downscaled simulated dataset.

There is significant spatial variation in the observed responses of both TP and TN during periods of drought, warm months and compound events (Fig. 6). The concentrations of TP are found to either increase or decrease during drought periods (Fig. 6a). For instance, in Europe, the median percentage change in TP due to drought is +8%. Whereas, in North America, the median percentage change in TP for drought periods is −3%. This result is overall consistent with the observed responses in the training and testing datasets of this study, which show both positive and negative relations between discharge and TP concentrations across different regions of the world (Extended Data Fig. 6-7). Furthermore, Asia exhibits an average increase in TP concentration of +11% during droughts, while Africa has an average change of −9% under drought. Similar mixed responses are found for TN during periods of drought (Fig. 6b). For example, the median change in TN during droughts is +8% in North America and −4% in Africa. Furthermore, Asia exhibits an average change in TN of +18% during droughts, while Australia has an average change in TN of −19% under drought. Furthermore, both positive and negative correlations are also found between discharge and TN concentrations in the training and testing datasets (Extended Data Fig. 6-7).

The spatial patterns with responses of TP concentrations show consistent increases during warm months with high water temperature throughout almost all regions of the world (Fig. 6c). For example, the global median change in TP concentrations during warm months is +21%. Furthermore, the median change in TP per region for periods of high water temperature are +20% (North America), +11% (South America), +21% (Europe), +14% (Africa), +31% (Asia) and +16% (Australia). . Furthermore, there is a relatively high positive correlation between TP and water temperature for a significant number of monitoring stations included in the training data (Extended Data Fig. 6-7). This is also likely a key reason for the observed increases in TP under periods of high temperature simulated by the hybrid process-based and machine learning monthly model. Compared to TP, warm months lead to more mixed responses in TN concentrations (Fig. 6d). . There is also a general negative correlation between water temperature and TN concentrations in the datasets used for training and testing (Extended Data Fig. 6-7), which further explains the simulated decrease in TN concentrations under warm months in most regions (Fig. 6b). During compound events of drought and high water temperature, TP tends to increase the most compared to TN concentrations (Fig. 6e,f), although the TP concentration responses are less consistent across regions than found for warm spells solely (Fig 6c).

There is a general increase in N:P ratio observed during droughts with a median change of +7% (Fig 7a). In contrast, periods of high water temperature and compound events show an overall decrease in N:P ratio (Fig. 7b-c) with a global median change in the N:P ratio during warm months and compound events of −11% and −4%, respectively. This result implies that the increases in TP tend to be stronger relative to TN during these hydro-climatic extremes.

We further analysed the regional time-series in TP and TN concentration over the period 1980-2010 (Fig. 8-9), and estimated the relative contribution of water temperature, discharge and precipitation on nutrient concentrations using feature importance analyses (Fig. 10). These time-series show significant positive trends ($p < 0.01$) in most regions of the world during 1980-2010. They also show the seasonal variability of TP and TN concentrations, which tends to coincide with discharge and water temperature (Extended Data Fig. 8-10). The feature importances associated with the random forest component of the hybrid framework for monthly TP concentration are: annual TP from IMAGE-GNM (0.271) followed by discharge (0.248), water temperature (0.241) and precipitation (0.240). Additionally, the feature importances for monthly TN concentrations are: 0.470 (Annual TN), 0.206 (Q) 0.173 (Tw) 0.149 (Precip). In both cases, the annual simulations from IMAGE-GNM are the variables with the greatest influence on the monthly concentrations obtained with downscaling. This is also highlighted by the partial dependence of each predictor variable on the simulated (monthly) TP and TN concentrations (Extended Data Fig. 11-12). In addition, the three hydrological variables (i.e. discharge, precipitation and water temperature) used in the random forest component each contribute significantly to the monthly variation in TP and TN concentration (Fig. 10, Extended Data Fig. 11-12).

4. Discussion and conclusion

In this study, we developed a hybrid modelling approach combining process-based nutrient modelling based on IMAGE-GNM with machine learning (random forest) resulting in a global monthly nutrient dataset (1980-2010), which was used to study the impacts of droughts, warm months and compound events on TP and TN concentrations. Our results show generally mixed responses of TP and TN concentrations during drought periods, while more consistent increases were found during warm months and compound events. In addition, we find phosphorus and nitrogen concentrations often follow similar seasonal patterns to water temperature and discharge.

The hybrid nutrient modelling framework developed in this study is useful to quantify the impacts of droughts and warm months both for regions with and without monitoring data available of TP and TN concentration globally. Furthermore, with this approach it is possible to understand the possible biases resulting from previous local-scale studies of the impacts of droughts and warm months on TP and TN concentrations. Most previous studies indicate that the responses of TP and TN to droughts and warm periods are highly varied across most locations worldwide (Macintosh et al., 2011; Hrdinka et al., 2012; Baurès et al., 2013). However, when training a hybrid approach globally based on the available data worldwide, it becomes clear that there is a strong relationship between drought-warm periods which may lead to widespread increases in total phosphorus concentrations in surface waters, which is

likely driven by lower dilution. Furthermore, reductions in nitrogen occur in many regions due to drought and warm periods likely driven by increased denitrification as well as lower pollutant mobilisation. This becomes clear when analysing the effects of these hydro-climatic extremes at large-scales using the hybrid nutrient model developed in this study and is not immediately apparent when studying these effects solely at the local scale.

During compound drought-warm events, the effects of lower dilution of phosphorus and increased denitrification appear to dominate. This leads to a general decrease in the N:P ratio during compound drought and warm months, which was previously less clear in local-scale studies (e.g. Zwolsman and van Bokhoven, 2007) but is elucidated at the global scale using this hybrid process-based and machine learning methodology.

Increased nutrient concentrations during periods of drought and high water temperature pose a significant risk of causing harmful algal blooms, which can lead to depletion of oxygen and adverse effects on freshwater ecosystems (Feng et al., 2024; Suresh et al., 2023). For instance, eutrophication caused by high nutrient concentrations can lead to mass die-offs of freshwater fish species, which can have long-lasting effects on the health of aquatic ecosystems (Zhou et al., 2024). In regions where nitrogen concentrations are abundant, increased concentrations of phosphorus can be a key driver of eutrophication and harmful algal blooms (Chen et al., 2023; Jarvie et al., 2018). This is fairly consistent with the results from our study, which show a general decrease in N:P ratios during warm months and during compound drought and warm months driven by stronger increases in TP compared to TN. However, increased TN concentrations during periods of drought can also be a key factor driving harmful algal blooms in surface waters (Chen et al., 2023; Jarvie et al., 2018). A general reduction in the N:P ratio driven by drought-warm months may lead to less phosphorus limitation, leading to either co-limitation of both nitrogen and phosphorus or a more nitrogen limited ecosystem. Over the coming decades, it is likely that the increased occurrence of droughts and heatwaves will lead to more frequent eutrophication in many regions of the world (Feng et al., 2024, Suresh et al., 2023). It is therefore crucial to develop water management strategies to reduce the potential impacts of increased eutrophication on freshwater ecosystems, as well as to maintain the usability of surface waters for domestic, industrial and agricultural uses (Wang et al., 2024; van Vliet et al., 2021).

A limitation of this study is the assumption that the downscaling approach, which is trained on monitoring station data, can be applied across most of the world at 0.5° x 0.5° spatial resolution. This could introduce challenges such as hydrological variables being outside the range of training data. However, we tried to mitigate this as far as possible by training the random forest component on a vast amount of data (1.1 million TP and 184,748 TN

concentration observations) and avoiding overfitting to data rich regions with stratified random sampling. Also, we focus on total phosphorous and nitrogen while the distinction between different nutrient forms in terms of particulate and dissolved forms (e.g. nitrate, nitrite, ammonium) are disregarded and may show rather opposite behaviour in responses during droughts. Another possible limitation of this study is the coarse spatial resolution (0.5° x 0.5°), which may reduce the ability to fully capture several local-scale processes affecting P and N in surface waters, such as pollutant dispersion or the exact location of several point sources of pollution. This instead leads to a more coarse analysis of P and N responses under droughts, warm months and compound extremes using the hybrid nutrient modelling framework developed in this study.

Despite these limitations, this study has demonstrated the strong potential of hybrid process-based and machine learning approaches to simulate nutrient concentrations in global surface waters. Furthermore, this study highlights the significant impacts of droughts and warm months on TP and TN concentrations worldwide during the period 1980-2010. Future directions could involve simulating different forms of nutrients (i.e. dissolved or particulate forms) using a similar hybrid methodology. Considering a warmer climate and potential increase in hydroclimatic extremes such as droughts and heatwaves, we conclude that there is an urgent need to address the potential impacts of droughts and warm months on phosphorus and nitrogen levels in global surface waters. To mitigate greater risk of eutrophication in many regions of the world, there is an urgent need for improved management strategies including a stricter regulation on nutrient pollution as well as improved monitoring of phosphorus and nitrogen concentrations during hydroclimatic extreme events.

5. Acknowledgement

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6. Data and code availability

Global monthly TP and TN concentration data from the hybrid process-based and machine learning model for 1980-2010 are available on Zenodo https://doi.org/10.5281/zenodo.19299931.

Python scripts associated with the hybrid model are available GitHub at https://github.com/SustainableWaterSystems/HybridNutrientModel/

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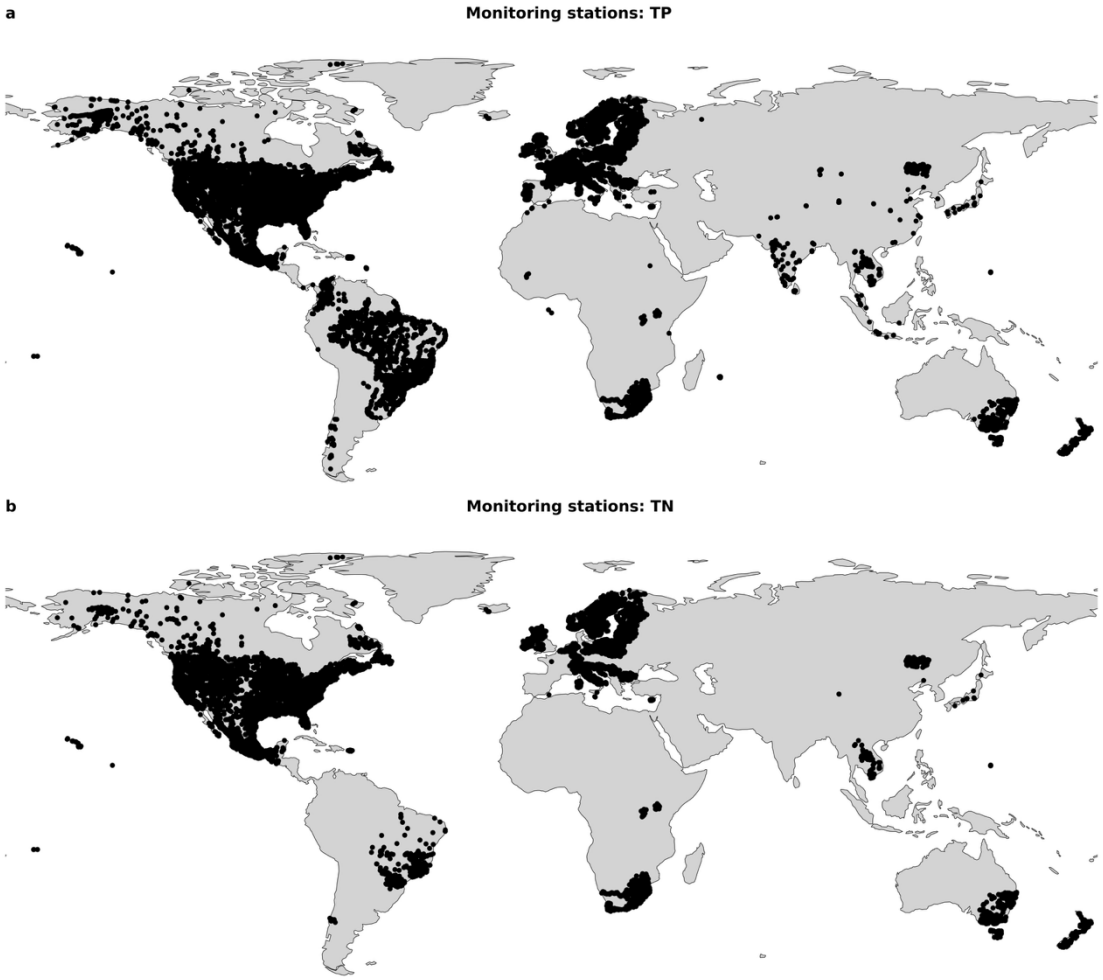
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Figure 1 - Locations with monitoring data of (**a**) total phosphorus (TP) concentration and (**b**) total nitrogen (TN) concentration. These data are originally from 7 water quality databases (United Nations Environment Programme, 2018; Hartmann et al., 2014; European Environment Agency, 2022; Government of Canada, 2021; Water Quality Portal, 2021;
765 Department of Water and Sanitation, 2022; National Institute of Water and Atmospheric Research, 2022), which were processed into a single dataset for TP and TN based on the procedure of Graham et al. (2024).

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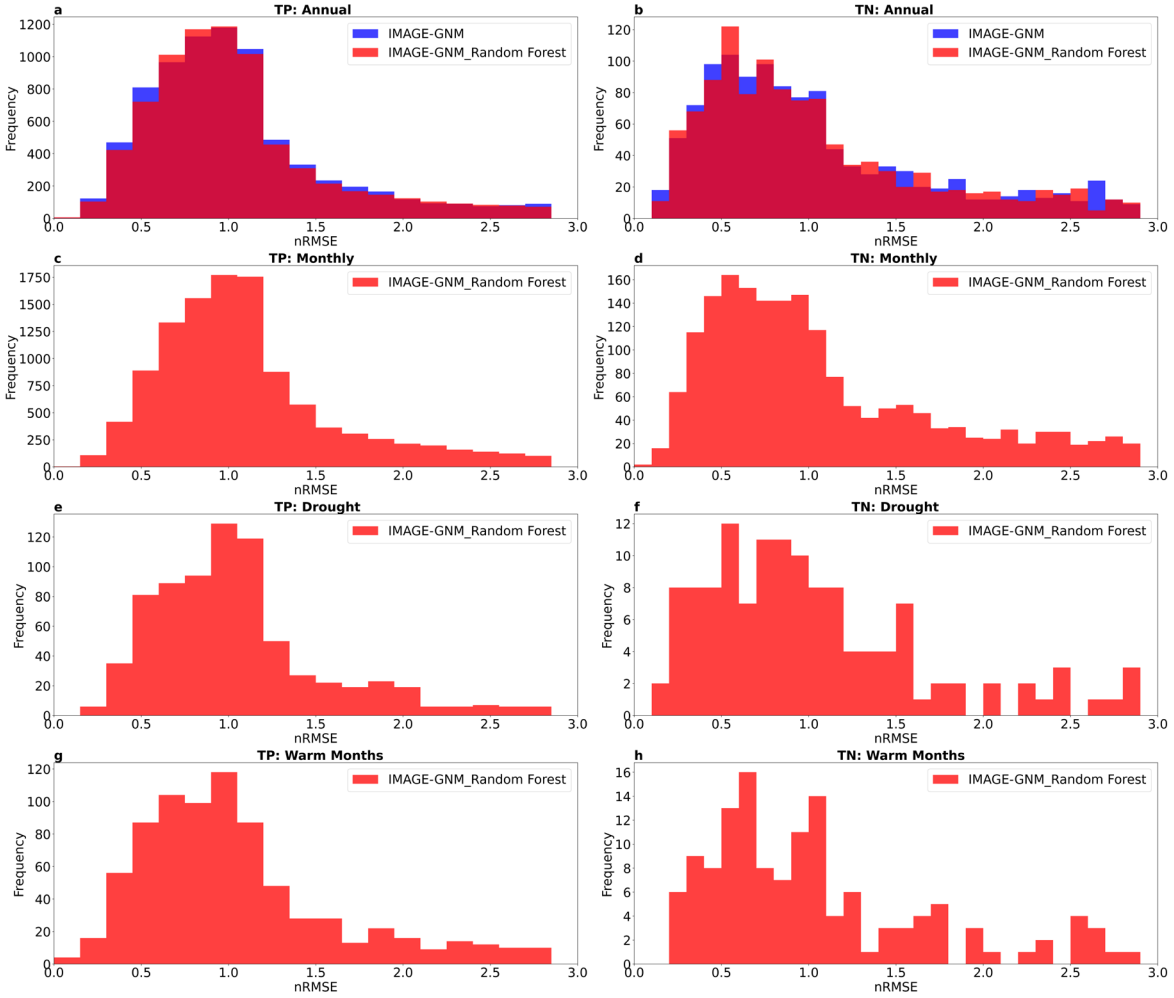


Figure 2 - Model validation based on the Root Mean Squared Error, normalised by the observed mean (nRMSE), for non-extreme and extreme conditions. (**a**) nRMSE of annual TP from IMAGE-GNM and the hybrid IMAGE-GNM_Random Forest model. (**b**) nRMSE of annual TN from IMAGE-GNM and IMAGE-GNM_Random Forest. (**c**) nRMSE of monthly TP from IMAGE-GNM_Random Forest. (**d**) nRMSE of monthly TN from IMAGE-GNM_Random Forest. (**e**) nRMSE of TP concentrations during drought months (**f**) nRMSE of TN concentrations during drought months (**g**) nRMSE of TP during warm months (**h**) nRMSE of TN during warm months. Y-axis: 'Frequency' represents the number of monitoring stations for 1980-2010.

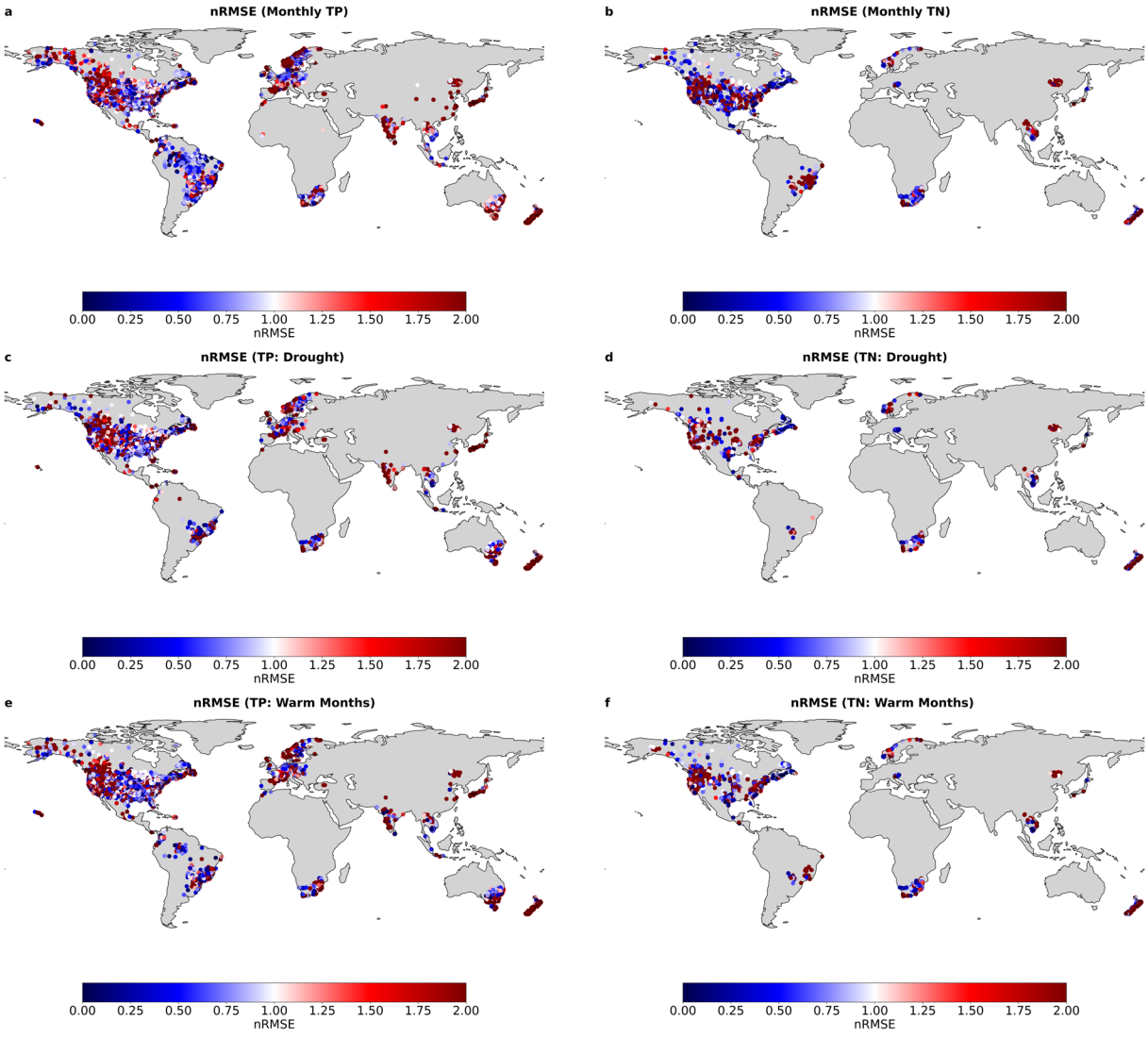


Figure 3 - Spatial distribution of normalised Root Mean Squared Errors (nRMSE) for non-extreme and extreme conditions. (**a**) nRMSE of monthly TP from IMAGE-GNM_Random Forest. (**b**) nRMSE of monthly TN from IMAGE-GNM_Random Forest (**c**) nRMSE of monthly TP during drought periods for IMAGE-GNM_Random Forest (**d**) nRMSE of monthly TN during drought periods for IMAGE-GNM_Random Forrest (**e**) nRMSE of monthly TP during warm months from IMAGE-GNM_Random Forest (**f**) nRMSE of monthly TN during warm months for IMAGE-GNM_Random Forest for 1980-2010.

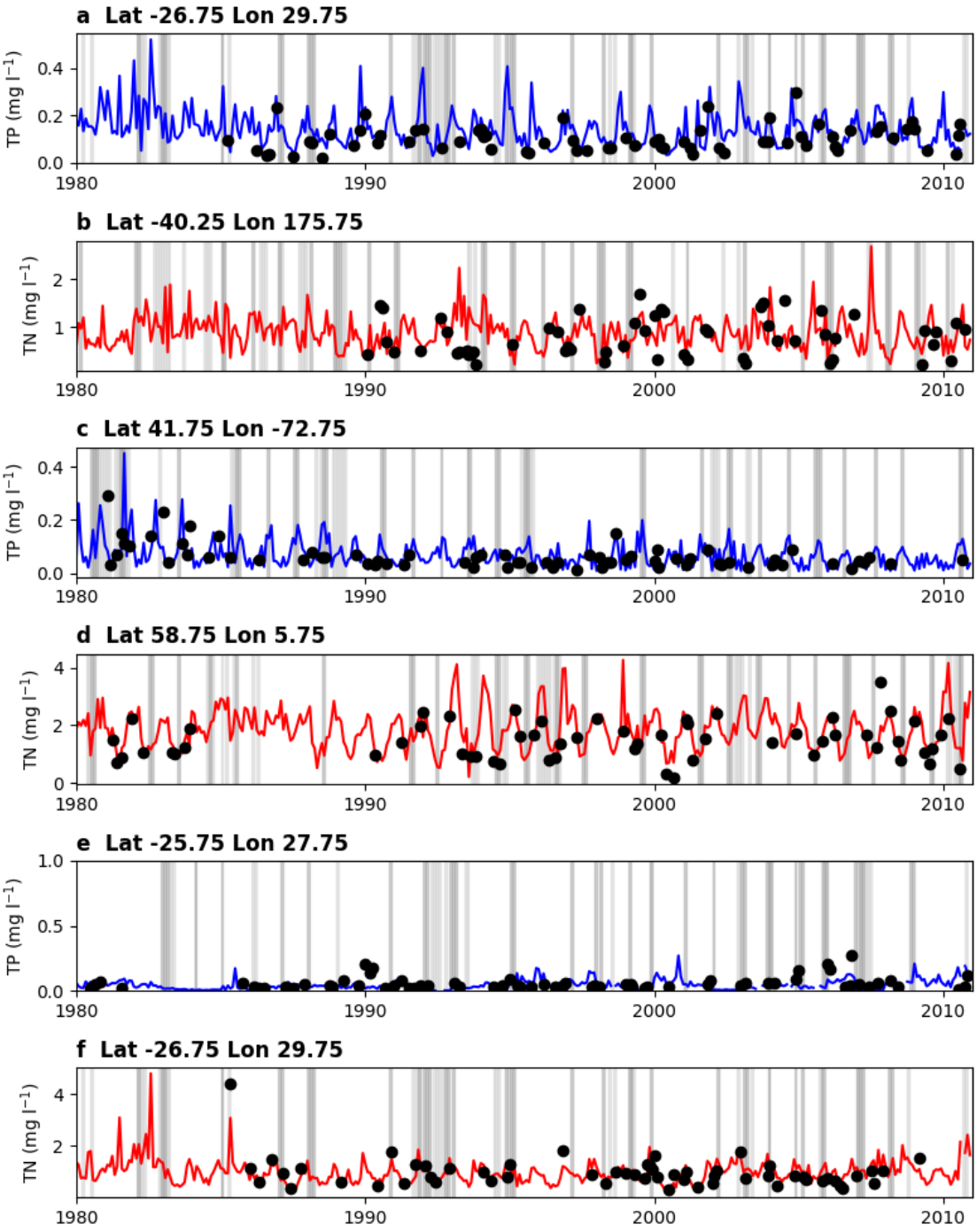


Figure 4 - Time-series of monthly TP and TN concentration simulated by the hybrid IMAGE-GNM_Random Forest model compared with monitoring data for multiple locations. Drought (grey). Warm month (dark grey).

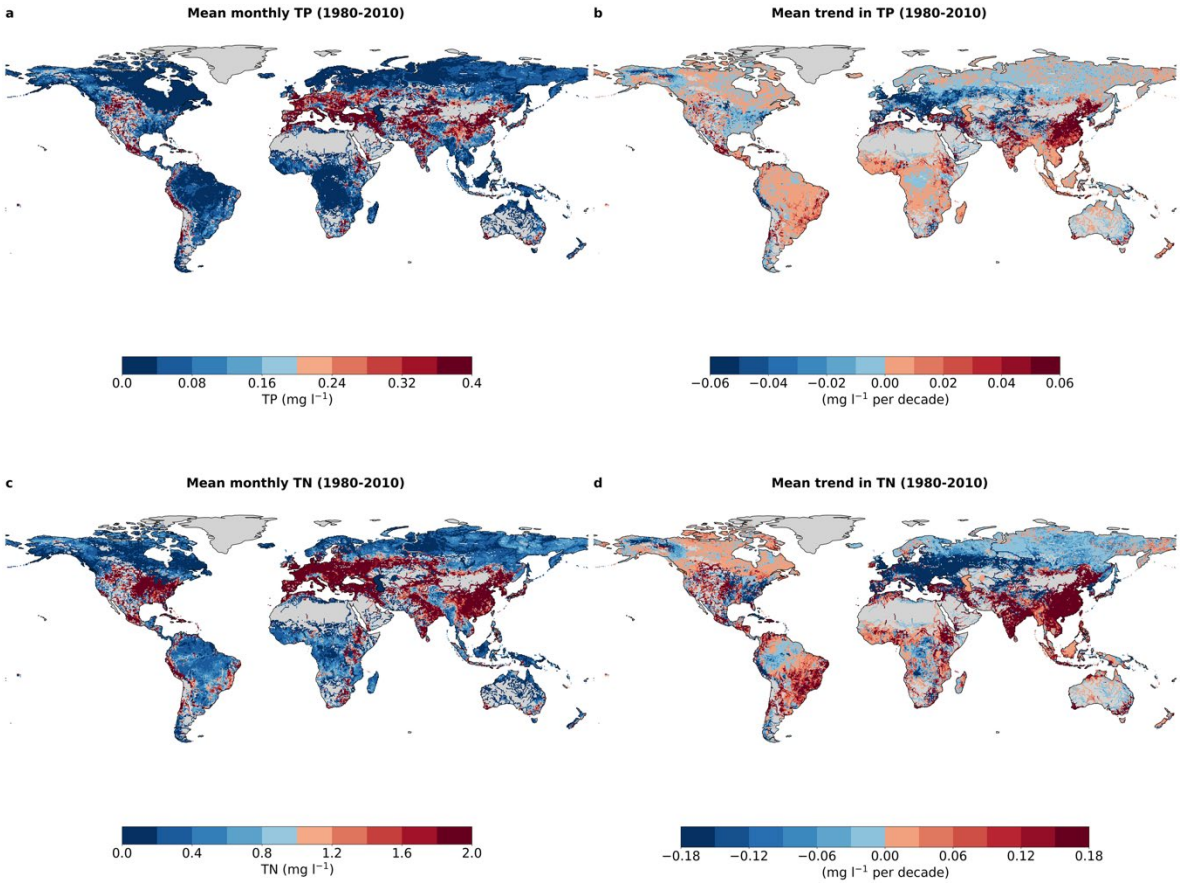
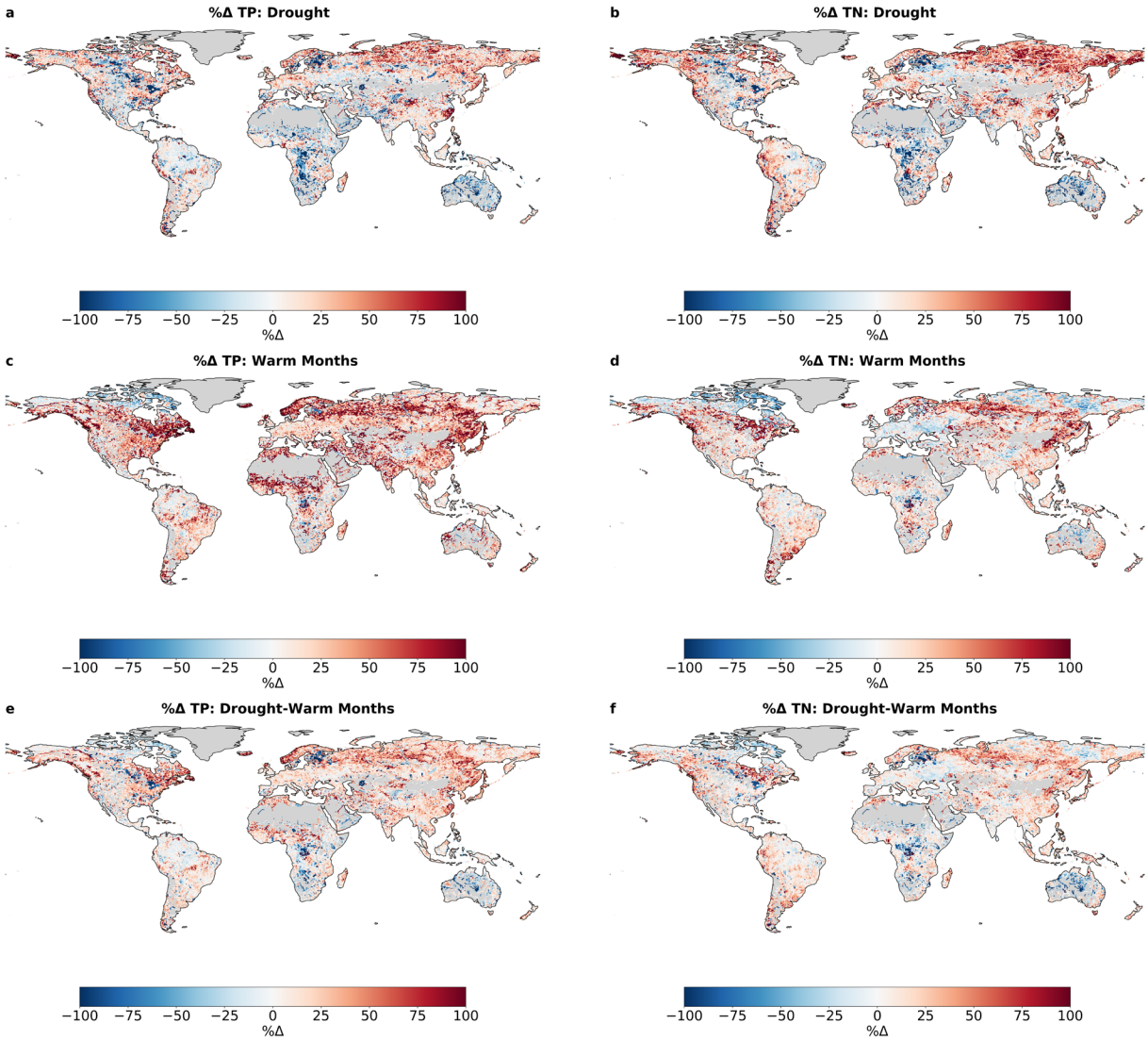
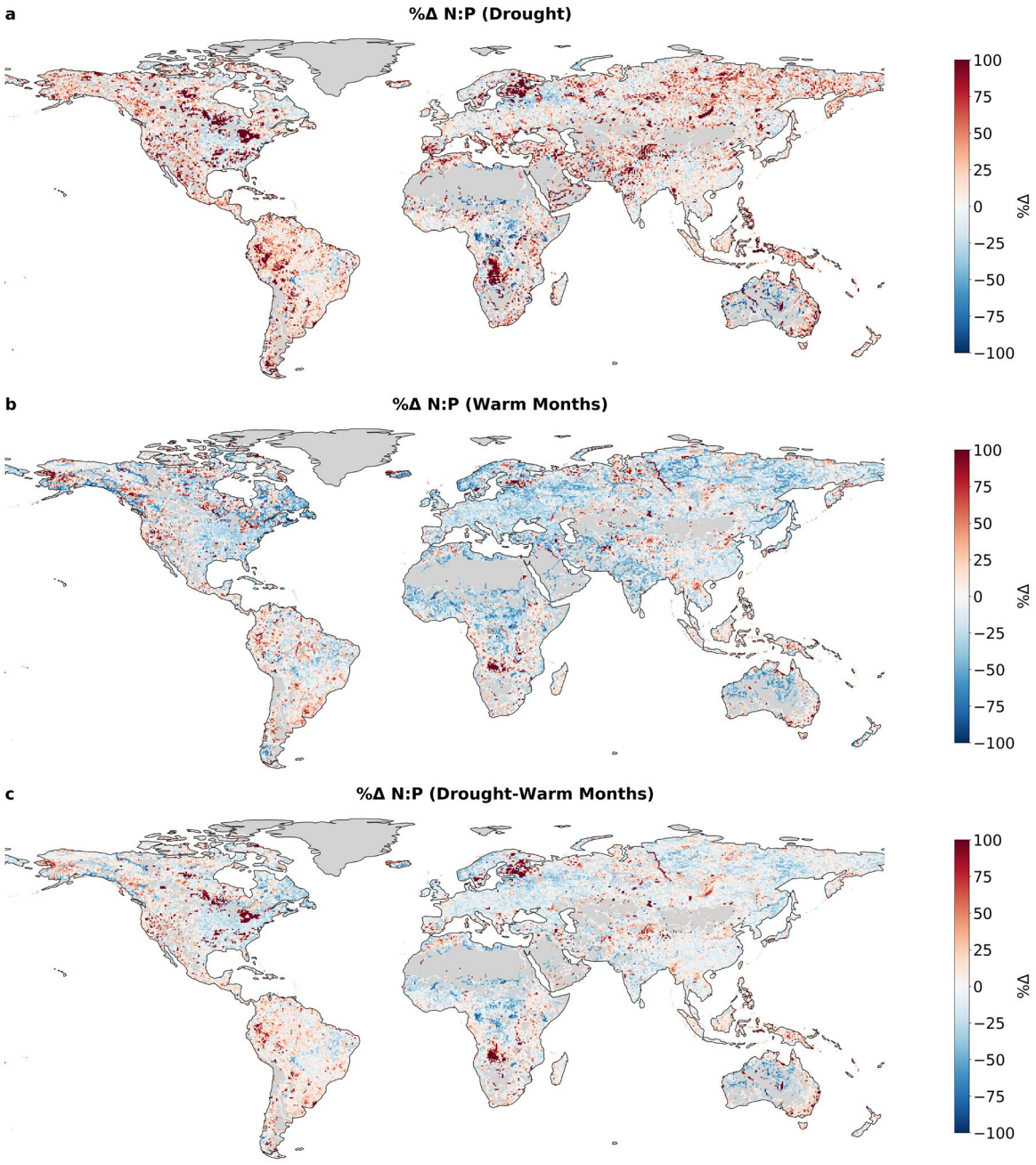


Figure 5 - Global patterns and trends in TP and TN concentration simulated by the downscaled monthly datasets. (**a**) mean monthly TP over the period 1980-2010. (**b**) average trend in TP during 1980-2010 (**c**) mean monthly TN over the period 1980-2010 (**d**) average trend in TN during 1980-2010.



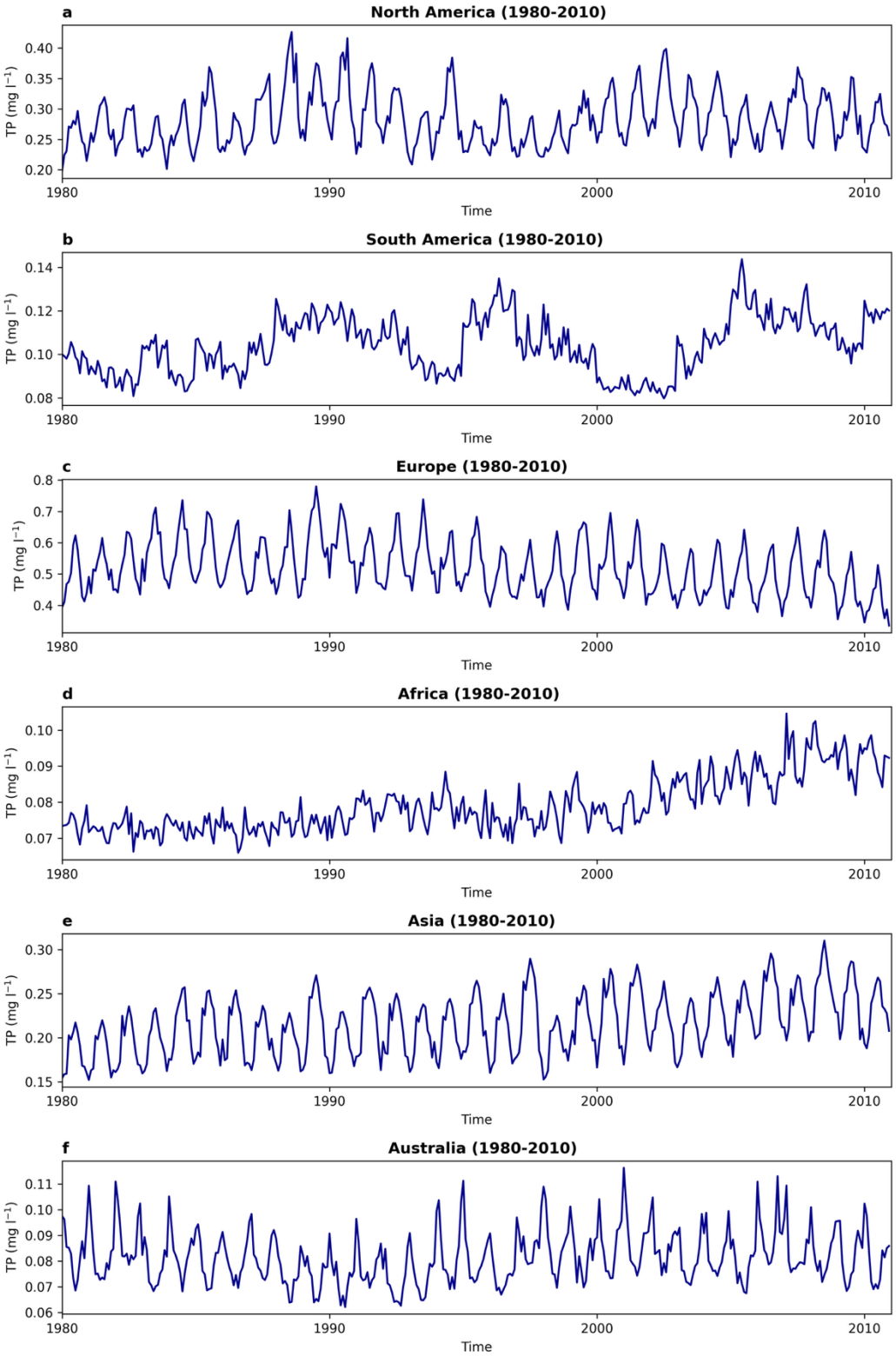
825 **Figure 6** – (**a**) Average percentage change in TP during droughts. (**b**) Average percentage change in TN during droughts. (**c**) Average percentage change in TP during warm months. (**d**) Average percentage change in TN during warm months. Average percentage change in TP during drought-warm months. (**e**) Average percentage change in TN during drought-warm months.

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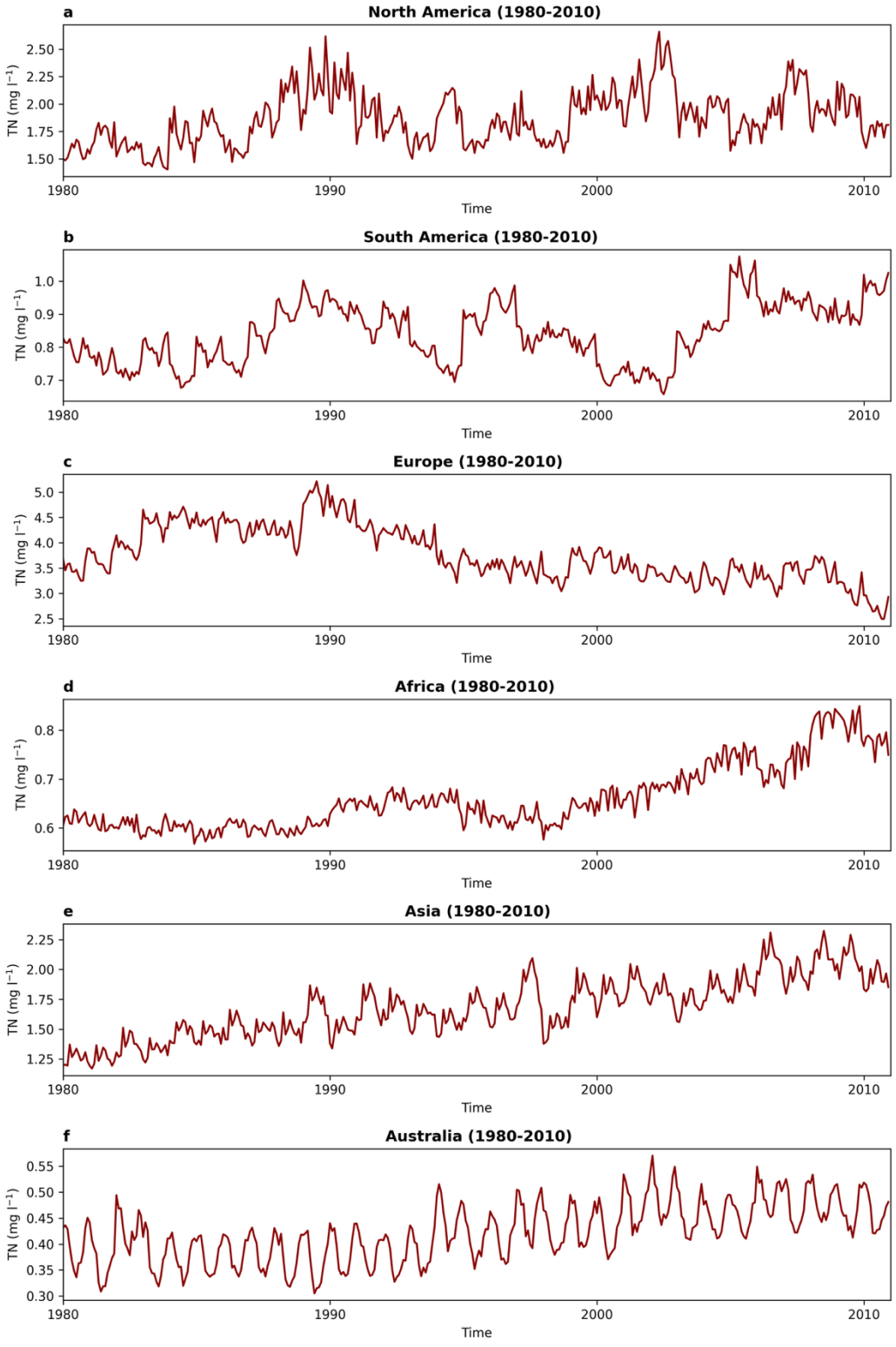
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Figure 7 - Average percentage change in N:P ratio during (**a**) droughts (**b**) warm months and (**c**) compound drought-warm months for 1980-2010.



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Figure 8 - Average monthly TP simulated by IMAGE-GNM_Random Forest per region.



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Figure 9 - Average monthly TN simulated by IMAGE-GNM_Random Forest per region.

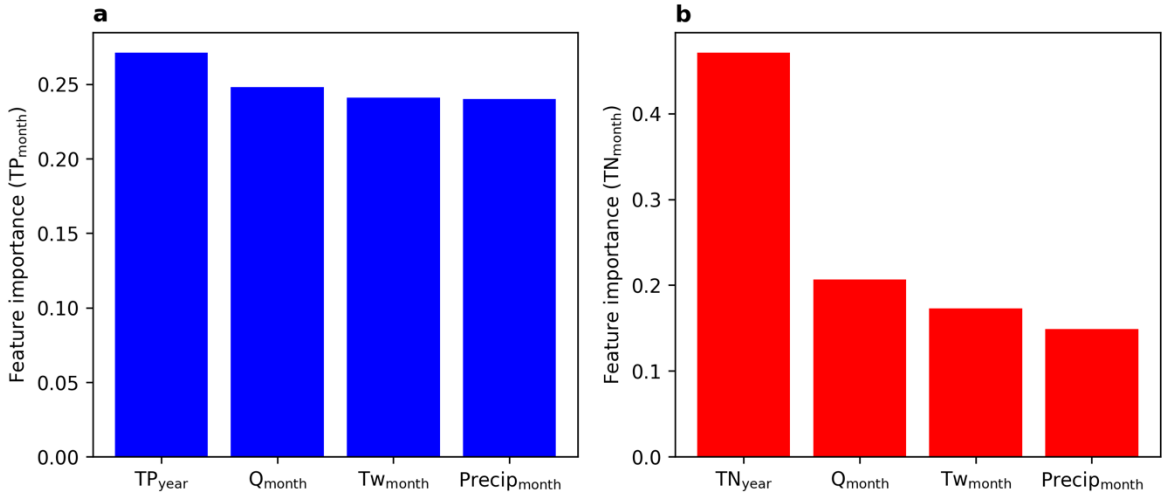
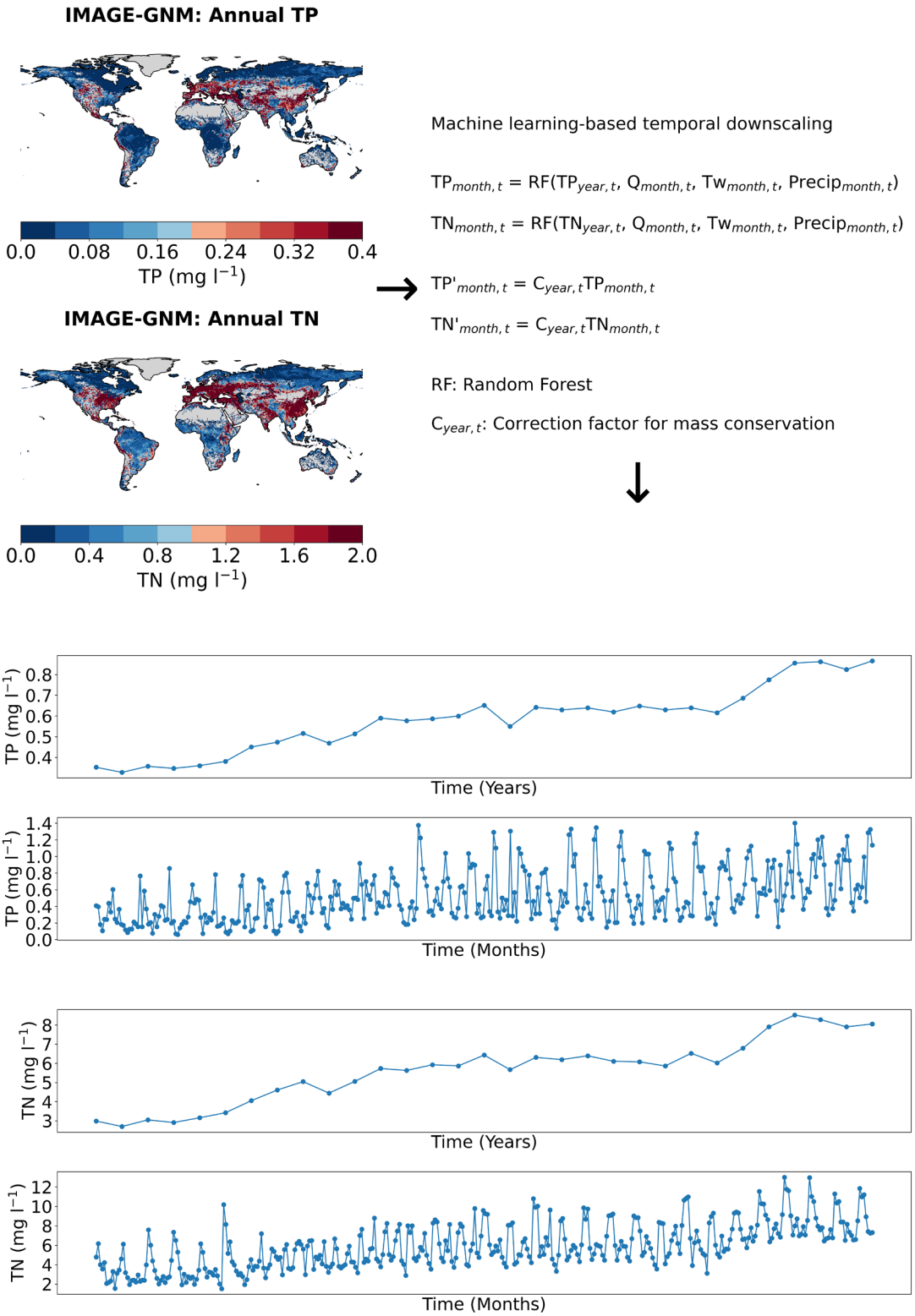
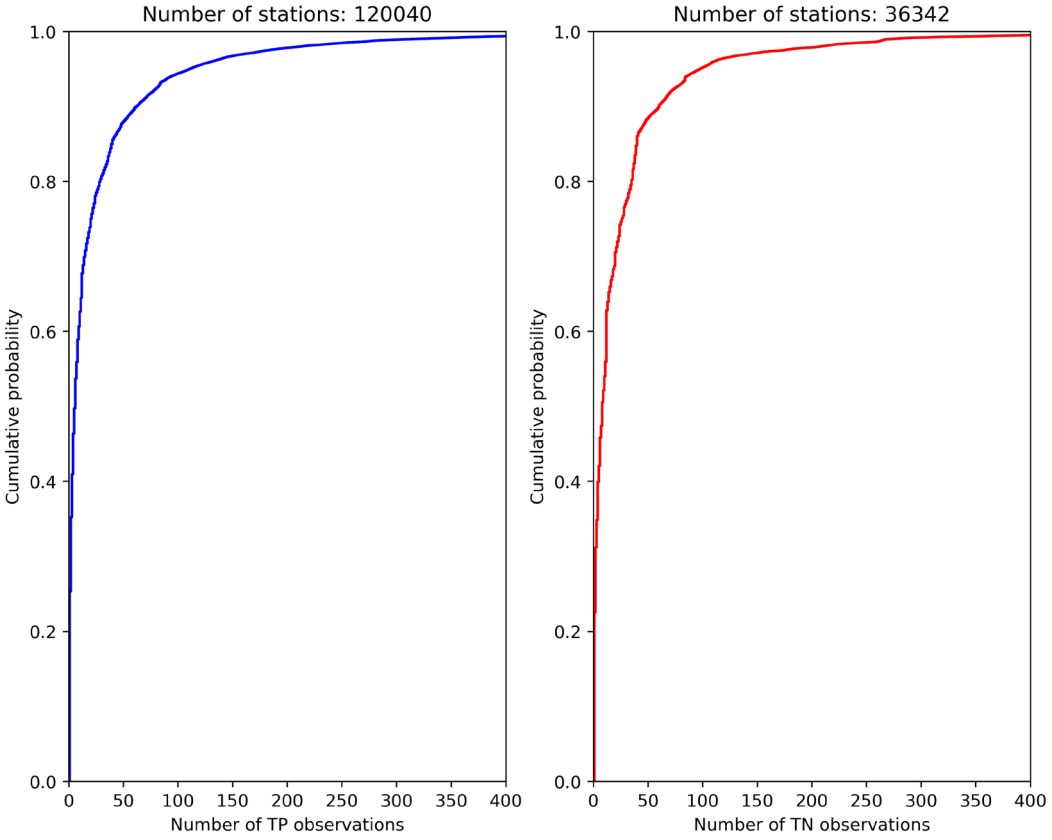


Figure 10 - Feature importances associated with the random forest component of the hybrid framework for (**a**) total phosphorus and (**b**) total nitrogen.

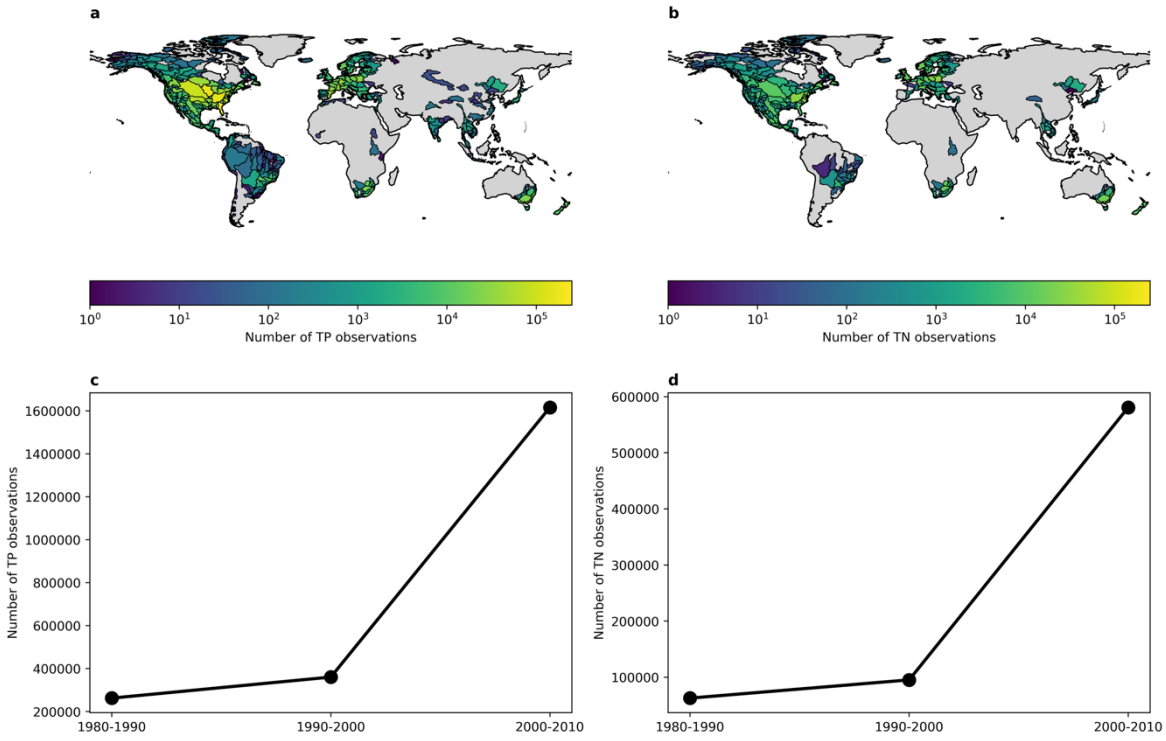


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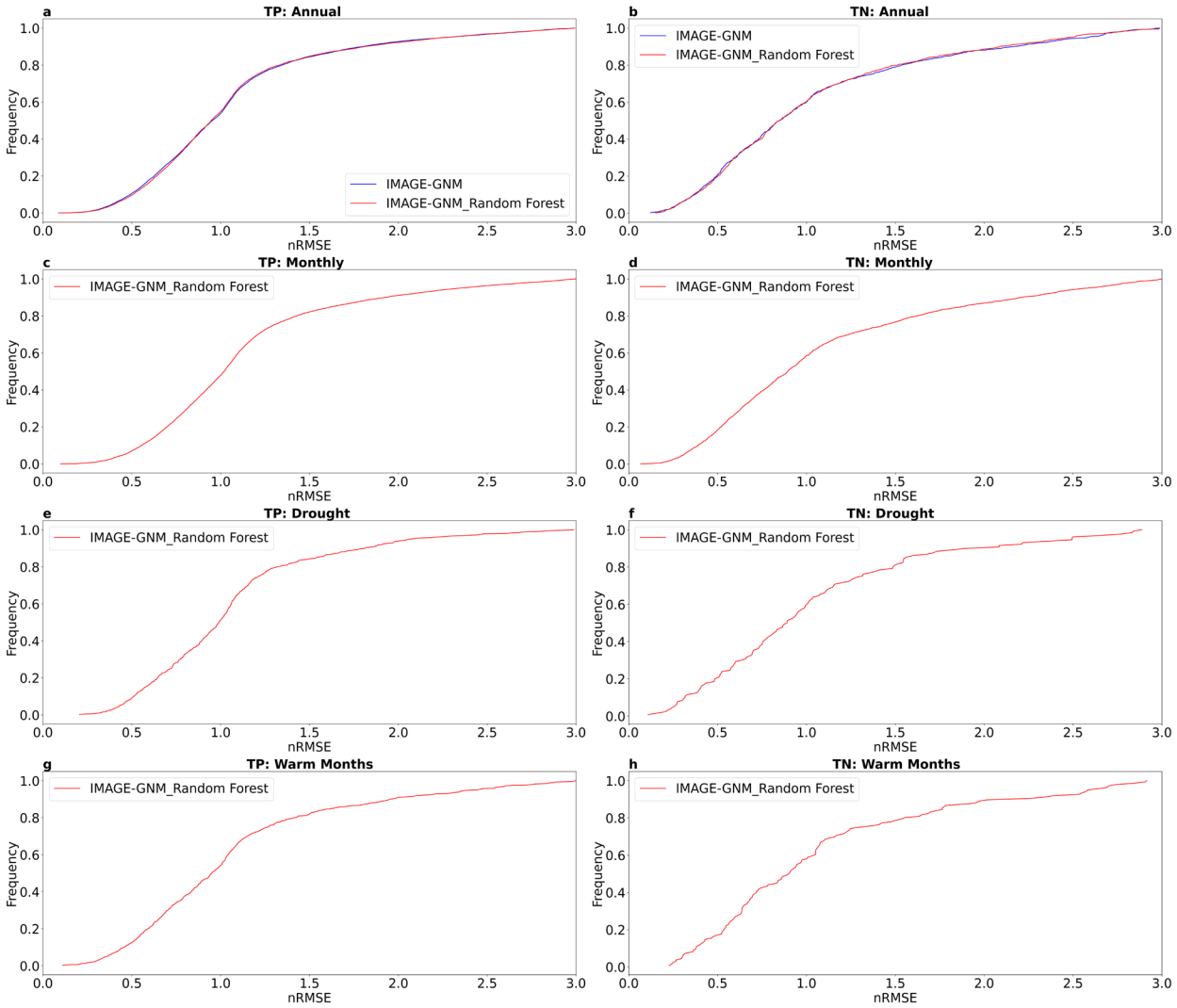
Extended Data Figure 1 - Hybrid nutrient modelling framework.



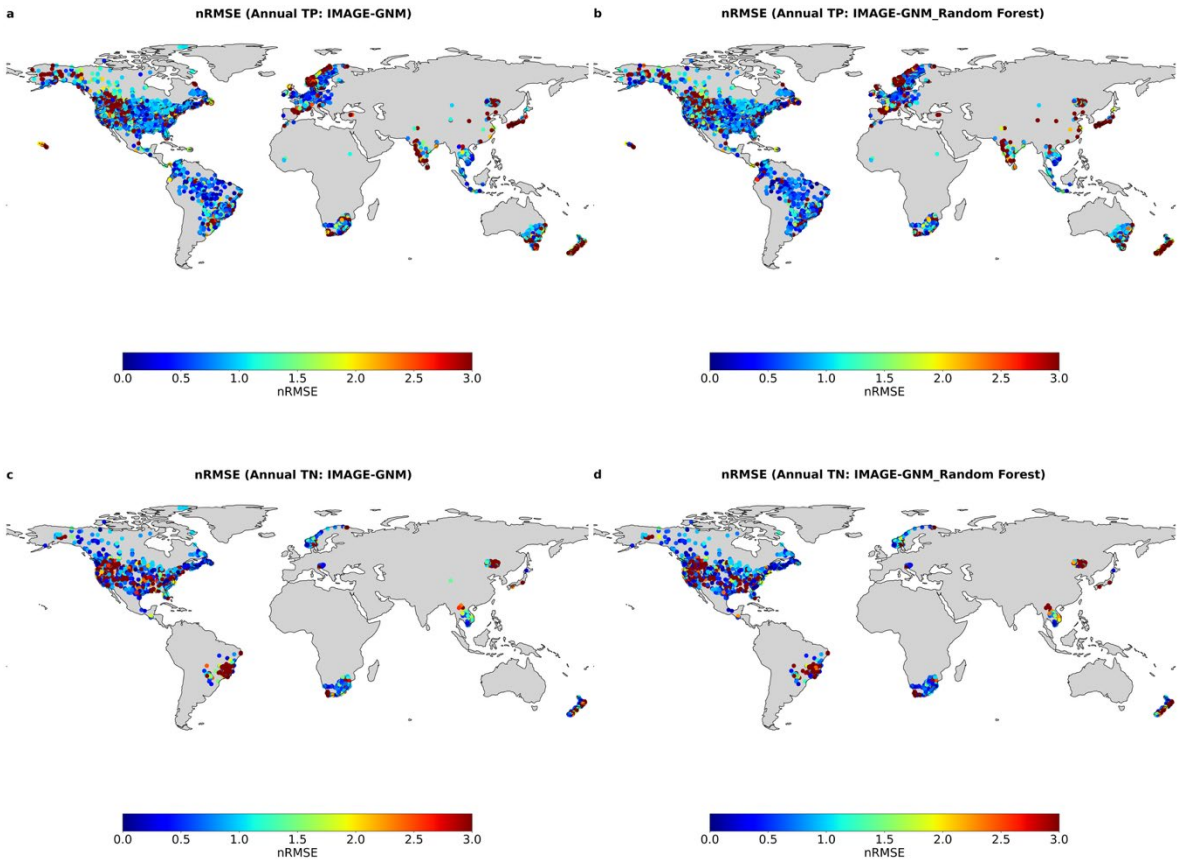
Extended Data Figure 2 - Cumulative distribution functions of the number of TP and TN observations per station.



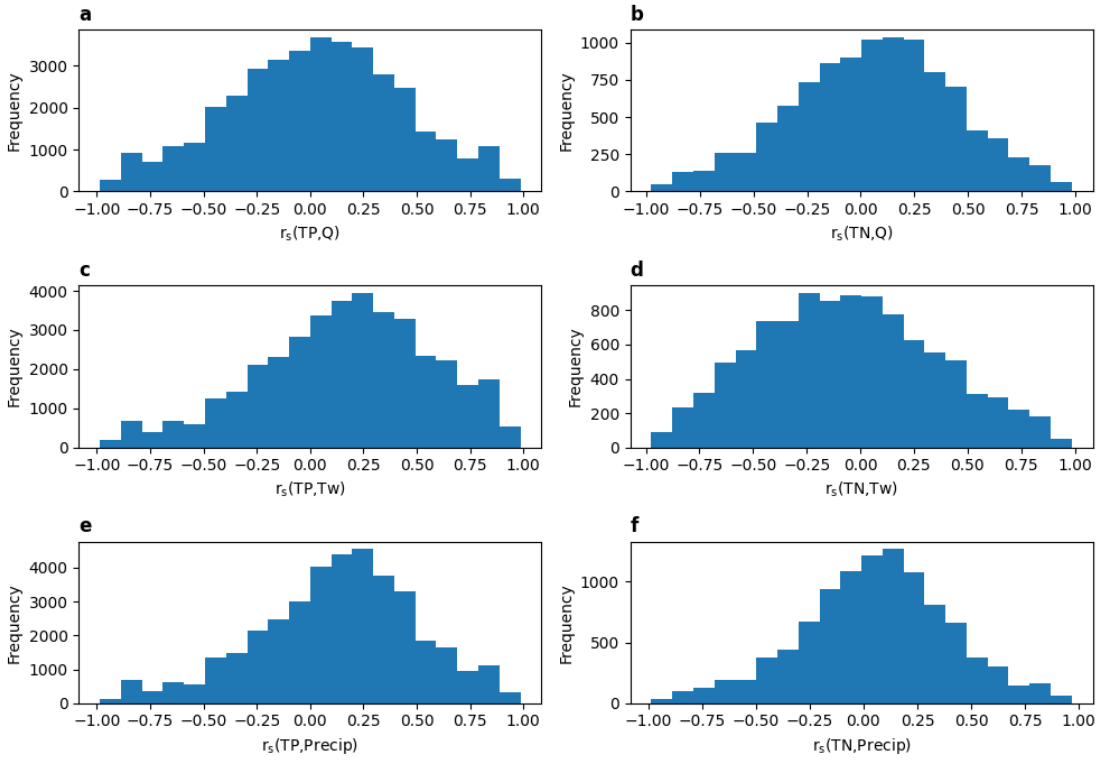
Extended Data Figure 3 - Number of observations per basin based on HydroBASINS level 4 used during stratified random sampling for (**a**) total phosphorus and (**b**) total nitrogen. Total number of observations per decade for (**c**) total phosphorus and (**d**) total nitrogen.



Extended Data Figure 4 - (**a**) CDF of nRMSE of annual TP from IMAGE-GNM and the hybrid IMAGE-GNM_Random Forest model. (**b**) CDF of nRMSE of annual TN from IMAGE-GNM and IMAGE-GNM_Random Forest. (**c**) CDF of nRMSE of monthly TP from IMAGE-GNM_Random Forest. (**d**) CDF of nRMSE of monthly TN from IMAGE-GNM_Random Forest. (**e**) CDF of nRMSE of TP concentrations during drought months (**f**) CDF of nRMSE of TN concentrations during drought months (**g**) CDF of nRMSE of TP during warm months (**h**) CDF of nRMSE of TN during warm months. Y-axis: 'Frequency' represents the cumulative probability. CDF: Cumulative distribution function.



Extended Data Figure 5 - Spatial distribution of annual-scale validation for IMAGE-GNM and IMAGE-GNM_Random Forest. (**a**) nRMSE of annual TP from IMAGE-GNM (**b**) nRMSE of annual TP from IMAGE-GNM_Random Forest (**c**) nRMSE of annual TN from IMAGE-GNM (**d**) nRMSE of annual TN from IMAGE-GNM_Random Forest.

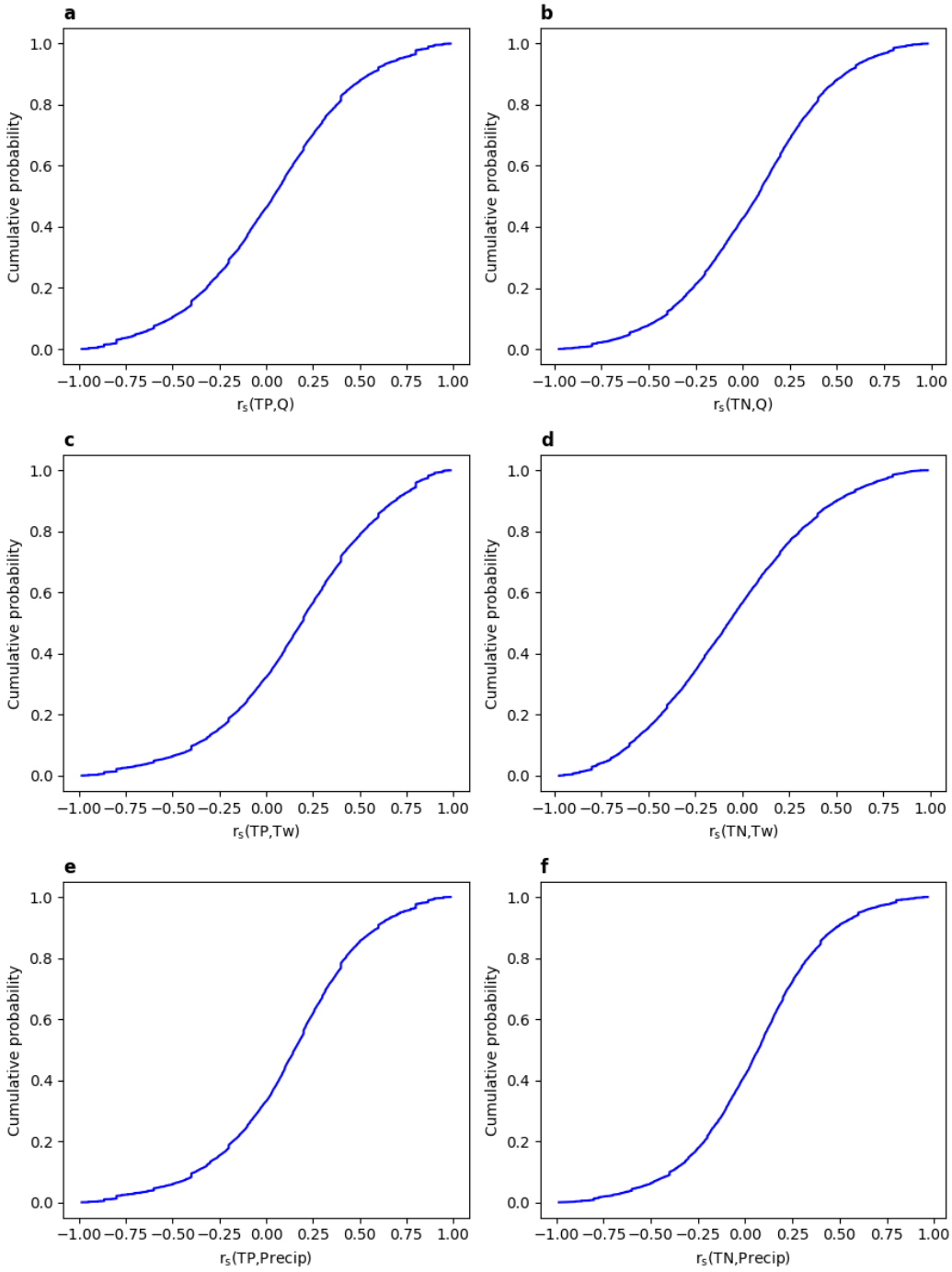


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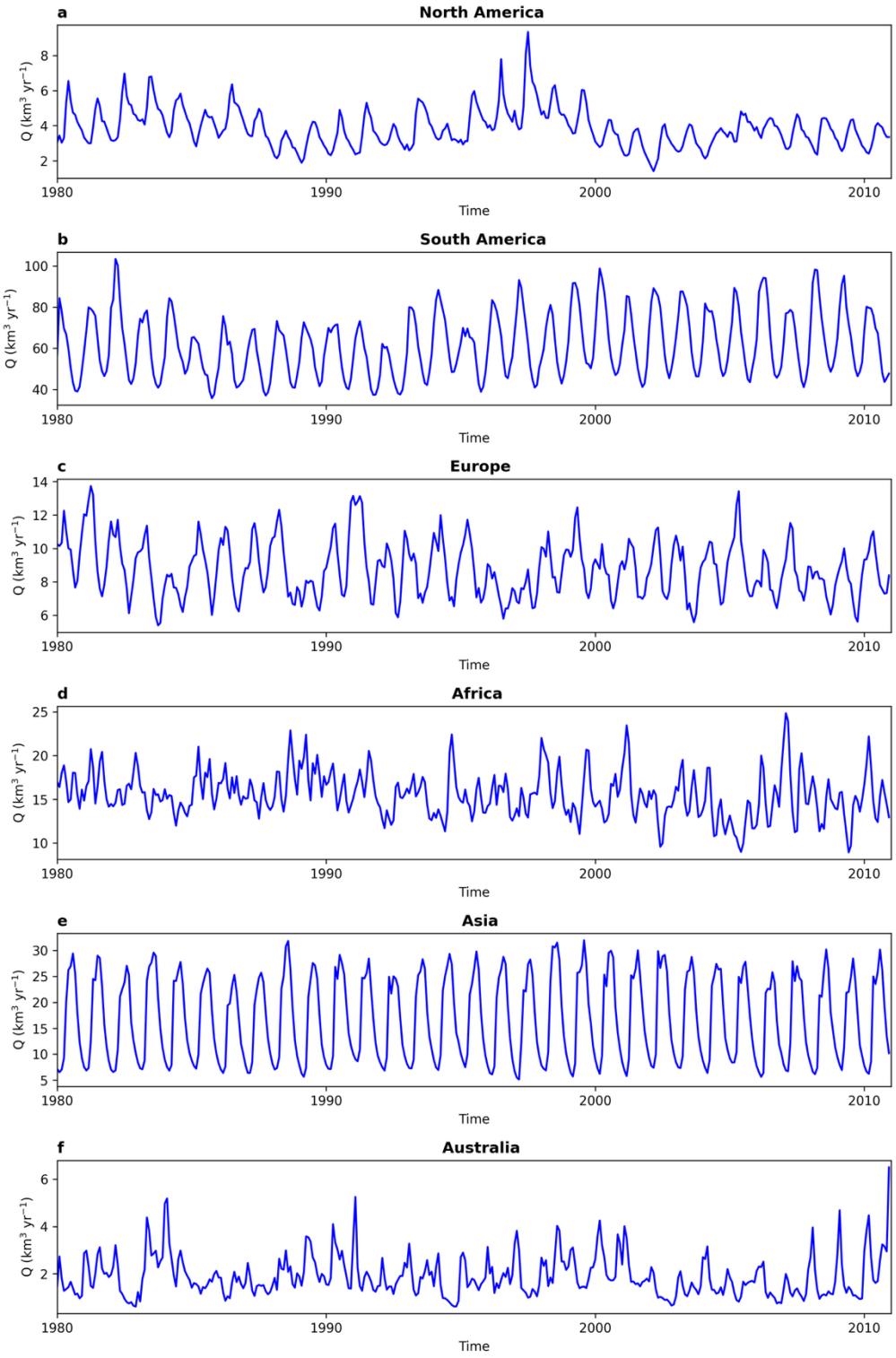
Extended Data Figure 6 - Histograms showing the spearman correlations between (**a**) in situ monitoring data of TP and simulated discharge (Q) (**b**) in situ monitoring data of TN and simulated Q (**c**) in situ monitoring data of TP and simulated water temperature (Tw) (**d**) in situ monitoring data of TN and simulated Tw (**e**) in situ monitoring data of TP and simulated precipitation (**f**) in situ monitoring data of TN and simulated precipitation.

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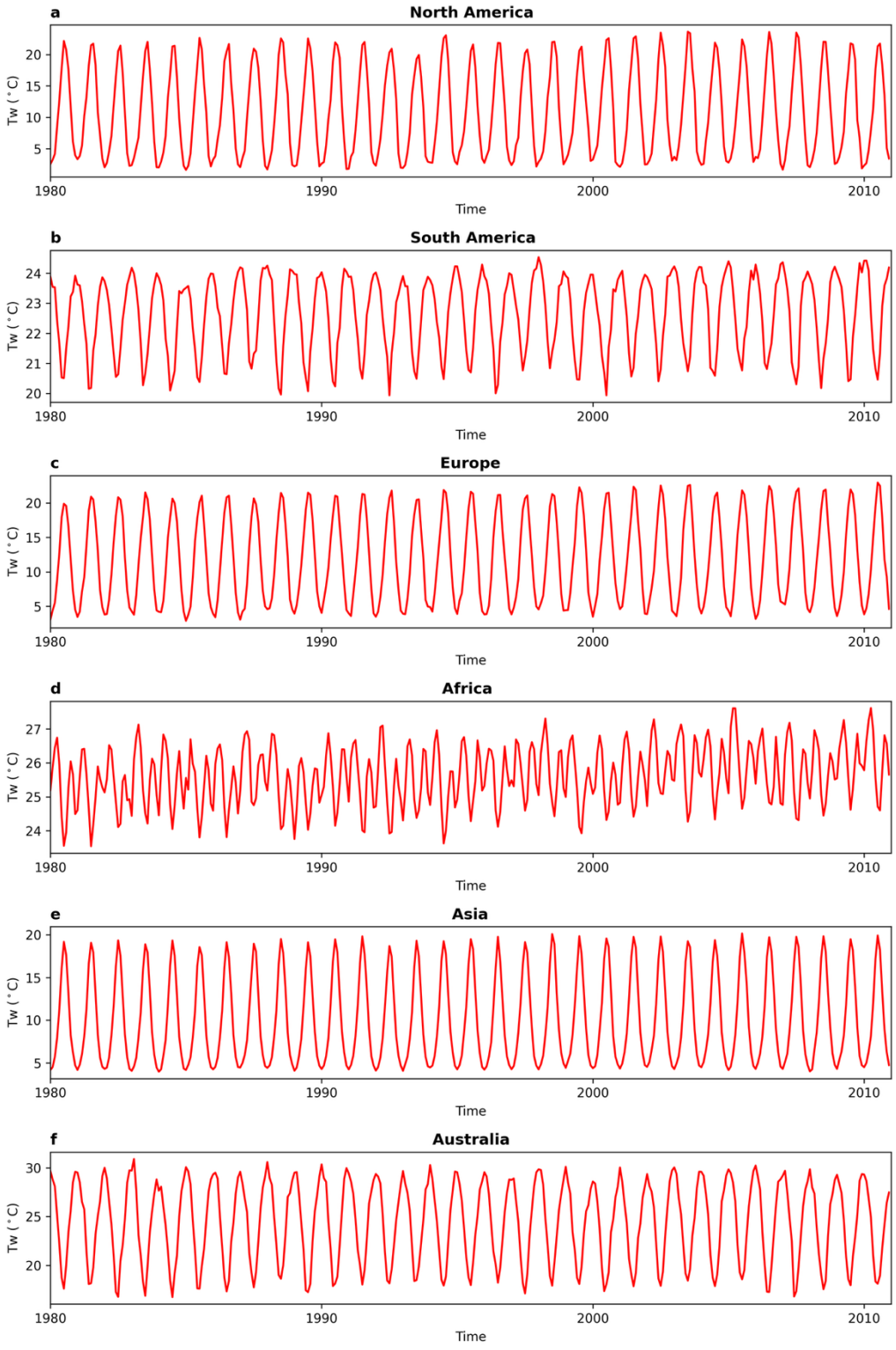


955 **Extended Data Figure 7** - CDFs showing the spearman correlations between (**a**) in situ monitoring data of TP and simulated discharge (Q) (**b**) in situ monitoring data of TN and simulated Q (**c**) in situ monitoring data of TP and simulated water temperature (Tw) (**d**) in situ monitoring data of TN and simulated Tw (**e**) in situ monitoring data of TP and simulated precipitation (**f**) in situ monitoring data of TN and simulated precipitation.



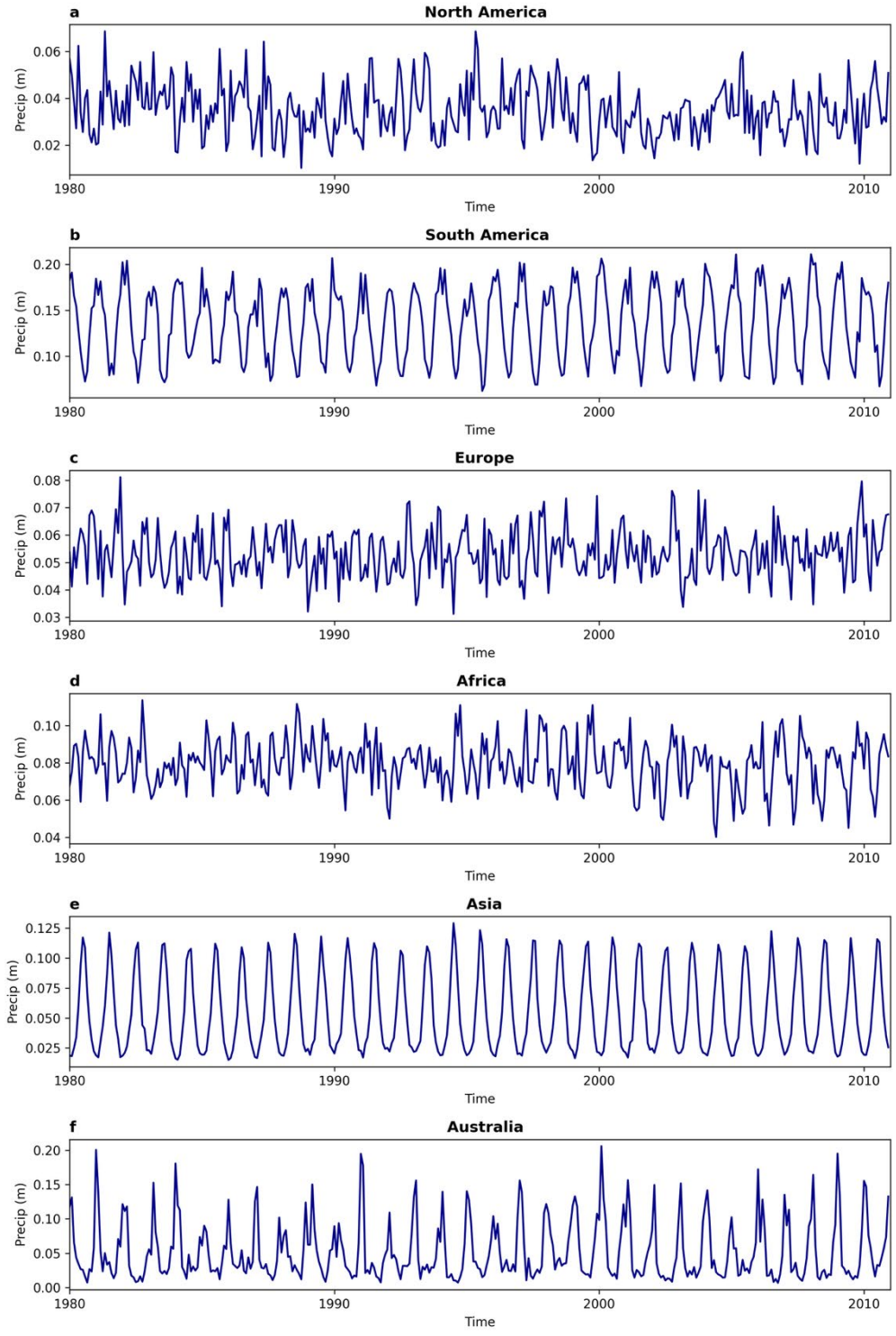
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Extended Data Figure 8 - Average monthly discharge per region over the period 1980-2010



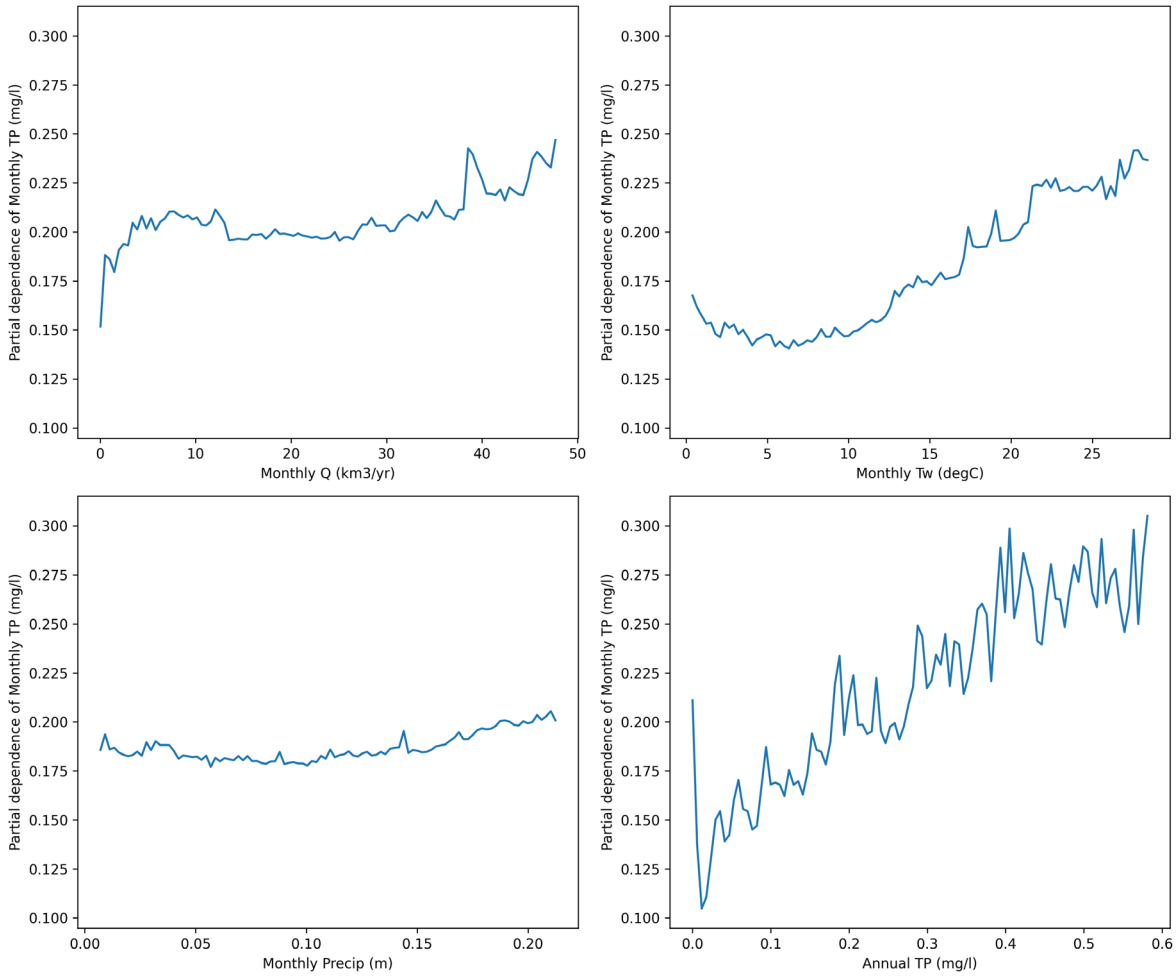
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Extended Data Figure 9 - Average monthly water temperature per region over the period 1980-2010

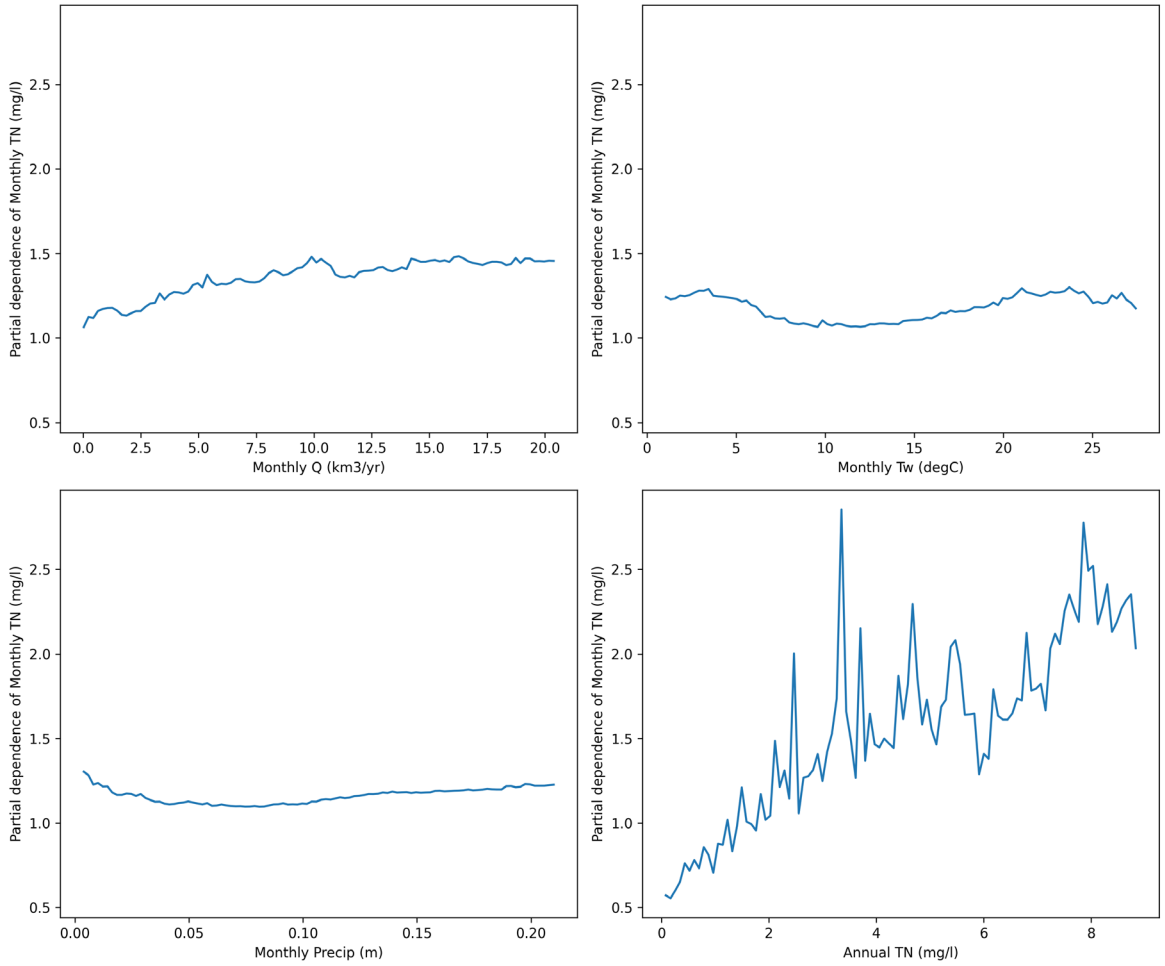


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Extended Data Figure 10 - Average monthly precipitation per region over the period 1980-2010.



Extended Data Figure 11 - Partial dependence of each predictor variable (Monthly Q, Monthly Tw, Monthly Precip, Annual TP) and its effect on target variable (Monthly TP).



Extended Data Figure 12 - Partial dependence of each predictor variable (Monthly Q, Monthly Tw, Monthly Precip, Annual TN) and its effect on target variable (Monthly TN).